

PRIMES IN ALMOST ALL SHORT INTERVALS II

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ABSTRACT. By combining the sieve machinery used in the author's previous paper on this topic and some new arithmetic information in Harman's monograph, the author proves that the interval $[n - n^{\frac{1}{22} + \varepsilon}, n]$ contains prime numbers for almost all n , improving the previous exponent $\frac{1}{21.5}$ by the author. The use of the variable role-reversal plays a crucial role in the proof.

CONTENTS

1. Introduction	1
2. An outline of the proof	2
3. Arithmetic Information I	3
4. Arithmetic Information II	18
5. Arithmetic Information III	18
6. Sieve asymptotic formulas	19
7. The final decomposition	23
8. Applications of Theorem 1.1	28
9. Remark	28
Appendix: Plot of Regions	29
References	30

1. INTRODUCTION

One of the famous topics in number theory is to find prime numbers in short intervals. In 1937, Cramér [7] conjectured that every interval $[n, n + f(n)(\log n)^2]$ contains prime numbers for some $f(n) \rightarrow 1$ as $n \rightarrow \infty$. The Riemann Hypothesis implies that for all large n , the interval $[n - n^\theta, n]$ contains $\sim n^\theta (\log n)^{-1}$ prime numbers for every $\frac{1}{2} < \theta \leq 1$. The first unconditional result of this asymptotic formula was proved by Hoheisel [18] in 1930 with $\theta = 1 - \frac{1}{33000}$. After the works of Hoheisel [18], Heilbronn [17], Chudakov [6], Ingham [20] and Montgomery [37], Huxley [19] proved in 1972 that the above asymptotic formula holds when $\theta > \frac{7}{12}$ by his zero density estimate. In 2026, Guth and Maynard [9] improved this to $\theta > \frac{17}{30}$ by a new zero density estimate.

In 1979, Iwaniec and Jutila [21] first introduced a sieve method into this problem. They established a lower bound with correct order of magnitude (instead of an asymptotic formula) with $\theta = \frac{13}{23}$. After that breakthrough, many improvements were made and the value of θ was reduced successively to

$$\begin{aligned} \frac{5}{9} \approx 0.5556, \quad \frac{11}{20} = 0.5500, \quad \frac{17}{31} \approx 0.5484, \quad \frac{23}{42} \approx 0.5476, \\ \frac{1051}{1920} \approx 0.5474, \quad \frac{35}{64} \approx 0.5469, \quad \frac{6}{11} \approx 0.5455 \quad \text{and} \quad \frac{7}{13} \approx 0.5385 \end{aligned}$$

by Iwaniec and Jutila [21], Heath-Brown and Iwaniec [16], Pintz [39] [40], Iwaniec and Pintz [22], Mozzochi [38] and Lou and Yao [32] [34] [35] [36] respectively. In 1996, Baker and Harman [2] presented an alternative approach to this problem. They used the alternative sieve developed by Harman [11] [12] to reduce θ to 0.535. Baker, Harman and Pintz [4] further developed this sieve process and combined it with Watt's theorem and showed $\theta = 0.525$. The best result now is due to the author [31], where he proved $\theta = 0.52$ is acceptable.

However, if we only consider the prime numbers in “almost all” intervals instead of “all” intervals, the intervals will be much shorter than $n^{0.52}$. In 1943, under the Riemann Hypothesis, Selberg [42] showed that Cramér's interval contains primes for almost all n if $f(n) \rightarrow \infty$ as $n \rightarrow \infty$. In the same paper, he also showed unconditionally that the interval $[n - n^{\frac{19}{77} + \varepsilon}, n]$ contains prime numbers for almost all n . In 1971, Montgomery [37] improved the exponent $\frac{19}{77}$ to $\frac{1}{5}$ with an asymptotic formula. The zero density estimate of Huxley [19] gives the exponent $\frac{1}{6}$ with an asymptotic formula, and the best asymptotic result now is also due to Guth and Maynard [9], where they proved the exponent $\frac{1}{7.5}$.

In 1982, Harman [10] used his alternative sieve method to showed that the interval $[n - n^{\frac{1}{10} + \varepsilon}, n]$ contains prime numbers for almost all n . His method can only provide a lower bound instead of an asymptotic formula. The exponent $\frac{1}{10}$ was reduced successively to

$$\frac{1}{12} \approx 0.0833, \frac{14}{159} \approx 0.0881, \frac{1}{13} \approx 0.0769, \frac{17}{227} \approx 0.0749, \frac{1}{13.5} \approx 0.0740, \\ \frac{1}{14} \approx 0.0714, \frac{1}{15} \approx 0.0667, \frac{1}{16} = 0.0625, \frac{1}{18} \approx 0.0556, \frac{1}{20} = 0.0500 \text{ and } \frac{1}{21.5} \approx 0.0465$$

by Harman [11] (and Heath-Brown [15]), Lou and Yao [33], Jia [24] [25], Lou and Yao [33], Li [28], Jia [23] (and Watt [44]), Li [29], Baker, Harman and Pintz [3], Wong [45] (and Jia [27], Harman [[14], Chapter 9]), Jia [26] and the author [30] respectively.

In the end of [30], the author mentioned that an exponent $\frac{1}{22}$ is far out of reach using the methods in [30]. Indeed, there are still some extra asymptotic regions proved in Harman's monograph [[14], Chapter 9] that we did not use in [30]. Fortunately, we find that one of those extra asymptotic regions is very powerful when combined with a variable role-reversal technique. In the present paper, we combine the sieve machinery used in [30] with the arithmetic information used in [[14], Chapter 9] to obtain the following result.

Theorem 1.1. *The interval $[n - n^{\frac{1}{22} + \varepsilon}, n]$ contains prime numbers for almost all n . Specifically, suppose that B is a sufficiently large positive constant, ε is a sufficiently small positive constant and X is sufficiently large. Then for positive integers $n \in [X, 2X]$, except for $O(X(\log X)^{-B})$ values, the interval $[n - n^{\frac{1}{22} + \varepsilon}, n]$ contains $\gg n^{1/22 + \varepsilon}(\log n)^{-1}$ prime numbers.*

Throughout this paper, we always suppose that B is a sufficiently large positive constant, ε is a sufficiently small positive constant, X is a sufficiently large integer, $x \in [X, 2X]$ and $K_0 > 0$. The appearance of K_0 in the exponent of a logarithm will always signify that the result holds for every $K_0 > 0$ with an implied constant that depends on K_0 . The letter p , with or without subscript, is reserved for prime numbers. Let c_0, c_1 and c_2 denote positive constants which may have different values at different places, and we write $m \sim M$ to mean that $c_1 M < m \leq c_2 M$. Let $b = 1 + \frac{1}{\log X}$, $\delta = \varepsilon^{1/3}$, $\eta = (2X)^{-\frac{21}{22} + \varepsilon}$, and $\eta_1 = \exp(-3(\log X)^{1/3})$. We use $M(s)$, $N(s)$, $H(s)$ and some other capital letters to denote divisor-bounded Dirichlet polynomials

$$M(s) = \sum_{m \sim M} a_m m^{-s}, \quad N(s) = \sum_{n \sim N} b_n n^{-s}, \quad H(s) = \sum_{h \sim H} c_h h^{-s},$$

and we use $L(s)$ to denote a “zeta-factor”

$$L(s) = \sum_{l \sim L} l^{-s}.$$

We say a Dirichlet polynomial $M(s)$ is *prime-factored* if we have

$$\left| M\left(\frac{1}{2} + it\right) \right| \ll M^{\frac{1}{2}}(\log x)^{-K_0}$$

for $\exp((\log x)^{1/3}) < |t| < x^{\frac{21}{22}}$. In fact, this holds when a_m is the characteristic function of primes or of numbers with a bounded number of prime factors restricted to certain ranges.

2. AN OUTLINE OF THE PROOF

Let $\mathcal{C}$ denote a finite set of positive integers, $p_j = X^{t_j}$ in the following sections and put

$$\mathcal{A} = \{a : a \in \mathbb{Z}, x - \eta x \leq a < x\}, \quad \mathcal{B}^1 = \{b : b \in \mathbb{Z}, x < b \leq 2x\}, \quad \mathcal{B}^2 = \{b : b \in \mathbb{Z}, x - \eta_1 x \leq b < x\},$$

$$\mathcal{C}_d = \{a : ad \in \mathcal{C}\}, \quad P(z) = \prod_{p < z} p, \quad S(\mathcal{C}, z) = \sum_{\substack{a \in \mathcal{C} \\ (a, P(z))=1}} 1.$$

Then we have

$$\pi(x) - \pi(x - \eta x) = S\left(\mathcal{A}, (2X)^{\frac{1}{2}}\right). \quad (1)$$

Buchstab's identity is the equation

$$S(\mathcal{C}, z) = S(\mathcal{C}, w) - \sum_{w \leq p < z} S(\mathcal{C}_p, p),$$

where $2 \leq w < z$.

In order to prove Theorem 1.1, we only need to show that $S\left(\mathcal{A}, (2X)^{\frac{1}{2}}\right) > 0$. By Buchstab's identity, we have

$$\begin{aligned} S\left(\mathcal{A}, (2X)^{\frac{1}{2}}\right) &= S\left(\mathcal{A}, X^{\frac{20}{209}}\right) - \sum_{\substack{\frac{20}{209} \leq t_1 < \frac{1}{2}}} S\left(\mathcal{A}_{p_1}, X^{\frac{20}{209}}\right) \\ &\quad + \sum_{\substack{\frac{20}{209} \leq t_1 < \frac{1}{2} \\ \frac{20}{209} \leq t_2 < \min(t_1, \frac{1}{2}(1-t_1))}} S(\mathcal{A}_{p_1 p_2}, p_2) \\ &= S_1 - S_2 + S_3. \end{aligned} \quad (2)$$

Our aim is to show that the sparser set $\mathcal{A}$ contains the expected proportion of primes compared to the larger sets $\mathcal{B}^i$, which requires us to decompose $S(\mathcal{A}, (2X)^{\frac{1}{2}})$ using the above Buchstab's identity, prove asymptotic formulas (for almost $x \in [X, 2X]$ except for $O(X(\log X)^{-B})$ values) of the forms

$$S(\mathcal{A}, z) = \eta(1 + o(1))S(\mathcal{B}^1, z) \quad \text{or} \quad S(\mathcal{A}, z) = \frac{\eta}{\eta_1}(1 + o(1))S(\mathcal{B}^2, z) \quad (3)$$

for some parts of it, and drop the remaining positive parts. We say a term $S(\mathcal{A}, z)$ has an asymptotic formula if any of the two forms of (3) holds for this term (or all of its decomposed parts).

In Sections 3 and 4 we provide some arithmetic information by proving mean value bounds for Dirichlet polynomials. In Section 5 we provide additional arithmetic information used in Harman's monograph [[14], Chapter 9]. We shall use them to prove asymptotic formulas for terms of the form $S(\mathcal{A}_{p_1 \dots p_n}, X^\delta)$ and $S(\mathcal{A}_{p_1 \dots p_n, p_n})$ in Section 6. In Section 7 we make further use of Buchstab's identity to decompose $S(\mathcal{A}, (2X)^{\frac{1}{2}})$ and prove Theorem 1.1.

3. ARITHMETIC INFORMATION I

In the following three sections we provide some arithmetic information which will help us prove the asymptotic formulas for sieve functions. In this section we only use the classical mean value estimate and Halász method.

Lemma 3.1. *Suppose that $MH = X$, $b_h = \Lambda(h)$ and $X^\delta \ll H \ll X^{\frac{20}{209}}$. Then for $(\log X)^{B/\varepsilon} \leq T \leq X$, we have*

$$\left(\min \left(\eta, \frac{1}{T} \right) \right)^2 \int_T^{2T} |M(b+it)H(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}.$$

Proof. Let $s = b + it$ and by the zero-free region of the ζ function, for $|t| \leq 2X$ we have

$$H(s) = \frac{(c_2 H)^{1-s} - (c_1 H)^{1-s}}{1-s} + O\left((\log x)^{-2B/\varepsilon}\right). \quad (4)$$

So, for $(\log X)^{B/\varepsilon} \leq |t| \leq 2X$ we have $H(s) \ll (\log x)^{-B/\varepsilon}$. According to the discussion in [16], there are $O((\log X)^2)$ sets $S(V, W)$, where $S(V, W)$ is the set of $t_k (k = 1, \dots, K)$ with the property $|t_r - t_s| \geq 1$ ($r \neq s$). Moreover,

$$V \leq M^{\frac{1}{2}} |M(b + it_k)| < 2V, \quad W \leq H^{\frac{1}{2}} |H(b + it_k)| < 2W,$$

where $X^{-1} \leq M^{-\frac{1}{2}}V, X^{-1} \leq H^{-\frac{1}{2}}W$ and $V \ll M^{\frac{1}{2}}, W \ll H^{\frac{1}{2}}(\log x)^{-B/\varepsilon}$. Then we have

$$\int_T^{2T} |M(b+it)H(b+it)|^2 dt \ll V^2 W^2 x^{-1} (\log x)^2 |S(V, W)| + O(x^{-2\varepsilon}),$$

where $S(V, W)$ is one of sets with the above properties.

Assume $X^{\frac{1}{k+1}} \leq H < X^{\frac{1}{k}}$, where k is a positive integer, $k \geq 10$ and $k\delta \ll 1$. Applying the mean value estimate to $M(s)$ and $H^k(s)$, we have

$$\begin{aligned} |S(V, W)| &\ll V^{-2}(M+T)(\log x)^d, \\ |S(V, W)| &\ll W^{-2k}(H^k+T)(\log x)^d, \end{aligned}$$

where $d = c/\delta^2$. Applying the Halász method to $M(s)$ and $H^k(s)$, we have

$$\begin{aligned} |S(V, W)| &\ll (V^{-2}M + V^{-6}MT)(\log x)^d, \\ |S(V, W)| &\ll (W^{-2k}H^k + W^{-6k}H^kT)(\log x)^d. \end{aligned}$$

Thus,

$$V^2 W^2 |S(V, W)| \ll V^2 W^2 F (\log x)^d,$$

where

$$F = \min \left\{ V^{-2}(M+T), V^{-2}M + V^{-6}MT, W^{-2k}(H^k+T), W^{-2k}H^k + W^{-6k}H^kT \right\}.$$

It will be proved that

$$\left(\min \left(\eta, \frac{1}{T} \right) \right)^2 V^2 W^2 F \ll \eta^2 x (\log x)^{-B/\varepsilon}.$$

We consider four cases.

(a) $F \leq 2V^{-2}M, F \leq 2W^{-2k}H^k$. Then

$$\begin{aligned} V^2 W^2 F &\ll V^2 W^2 \min \left\{ V^{-2}M, W^{-2k}H^k \right\} \\ &\leq V^2 W^2 (V^{-2}M)^{1-\frac{1}{2k}} (W^{-2k}H^k)^{\frac{1}{2k}} \\ &= V^{\frac{1}{k}} W M^{1-\frac{1}{2k}} H^{\frac{1}{2}} \\ &\ll x (\log x)^{-B/\varepsilon} \end{aligned}$$

and so

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 V^2 W^2 F \ll \eta^2 x (\log x)^{-B/\varepsilon}.$$

(b) $F > 2V^{-2}M, F > 2W^{-2k}H^k$. Then

$$\begin{aligned} V^2 W^2 F &\ll V^2 W^2 \min\{V^{-2}T, V^{-6}MT, W^{-2k}T, W^{-6k}H^k T\} \\ &\leq V^2 W^2 (V^{-2})^{1-\frac{3}{2k}} (V^{-6}M)^{\frac{1}{2k}} (W^{-2k})^{\frac{1}{k}} T \\ &= M^{\frac{1}{2k}} T. \end{aligned}$$

Since $k \geq 10$, we have $H \geq X^{\frac{1}{k+1}} \geq X^{1-\frac{k}{11}}, M^{\frac{1}{2k}} \ll X^{\frac{1}{22}}$, and so

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 V^2 W^2 F \ll \frac{\eta}{T} x^{\frac{1}{22}} T \ll \eta^2 x^{1-\varepsilon^2}.$$

(c) $F \leq 2V^{-2}M, F > 2W^{-2k}H^k$. Then

$$\begin{aligned} V^2 W^2 F &\ll V^2 W^2 \min\{V^{-2}M, W^{-2k}T, W^{-6k}H^k T\} \\ &\leq V^2 W^2 (V^{-2}M)^{1-\frac{1}{3k}} (W^{-6k}H^k T)^{\frac{1}{3k}} \\ &\ll MH^{\frac{1}{3}} T^{\frac{1}{3k}}, \end{aligned}$$

since $V \ll M^{\frac{1}{2}}$. As $H \geq X^{\frac{1}{k+1}} \geq X^{\frac{21}{44k}+\varepsilon}$ and $M \leq X^{1-\frac{21}{44k}}$, we have

$$\begin{aligned} \left(\min\left(\eta, \frac{1}{T}\right)\right)^2 V^2 W^2 F &\ll \eta^{2-\frac{1}{3k}} \left(\frac{1}{T}\right)^{\frac{1}{3k}} (MH)^{\frac{1}{3}} M^{\frac{2}{3}} T^{\frac{1}{3k}} \\ &\ll x^{\frac{1}{3}-\frac{21}{22}(2-\frac{1}{3k})+\frac{2}{3}(1-\frac{21}{44k})} \\ &\ll \eta^2 x^{1-\varepsilon^2}. \end{aligned}$$

(d) $F > 2V^{-2}M, F \leq 2W^{-2k}H^k$. Then

$$\begin{aligned} V^2 W^2 F &\ll V^2 W^2 \min\{V^{-2}T, V^{-6}MT, W^{-2k}H^k\} \\ &\leq V^2 W^2 (V^{-2}T)^{1-\frac{3}{2k}} (V^{-6}MT)^{\frac{1}{2k}} (W^{-2k}H^k)^{\frac{1}{k}} \\ &= M^{\frac{1}{2k}} HT^{1-\frac{1}{k}}. \end{aligned}$$

If $k \geq 11$, then $H \leq X^{\frac{1}{k}} \leq X^{1-\frac{21(k-1)}{11(2k-1)}}, M \gg X^{\frac{21(k-1)}{11(2k-1)}}$, and so

$$\begin{aligned} \left(\min\left(\eta, \frac{1}{T}\right)\right)^2 V^2 W^2 F &\ll \eta^{1+\frac{1}{k}} \left(\frac{1}{T}\right)^{1-\frac{1}{k}} (MH) M^{-\frac{2k-1}{2k}} T^{1-\frac{1}{k}} \\ &\ll x^{1-\frac{21}{22}(1+\frac{1}{k})-\frac{21(k-1)}{22k}} \\ &\ll \eta^2 x^{1-\varepsilon^2}. \end{aligned}$$

If $k = 10$, then $X^{\frac{1}{11}} \leq H \ll X^{\frac{20}{209}}, M \gg X^{\frac{189}{209}}$, and so

$$\begin{aligned} \left(\min\left(\eta, \frac{1}{T}\right)\right)^2 V^2 W^2 F &\ll \eta^{\frac{11}{10}} \left(\frac{1}{T}\right)^{\frac{9}{10}} (MH) M^{-\frac{19}{20}} T^{\frac{9}{10}} \\ &\ll x^{1-\frac{11}{10}\cdot\frac{21}{22}-\frac{19}{20}\cdot\frac{189}{209}} \\ &\ll \eta^2 x^{1-\varepsilon^2}. \end{aligned}$$

Combining the above cases, Lemma 3.1 is proved. $\square$

Lemma 3.2. Suppose that $MHK = X$, and M and H satisfy one of the following 9 conditions:

- (1) $MH \ll X^{\frac{172}{319}}, X^{\frac{21}{121}} \ll H, M^{29}H^{-1} \ll X^{\frac{109}{11}}, X^{\frac{63}{209}} \ll M, M^{-1}H^{29} \ll X^{\frac{67}{11}}, X^{\frac{63}{11}} \ll M^{12}H^{11};$
- (2) $MH \ll X^{\frac{19}{33}}, M^{29}H^{19} \ll X^{\frac{323}{22}}, X^{\frac{28}{33}} \ll M^2H, M^2H^{11} \ll X^{\frac{30}{11}}, X^{\frac{315}{11}} \ll M^{58}H^{49};$
- (3) $MH \ll X^{\frac{137}{242}}, X^{\frac{21}{110}} \ll H, M^6H \ll X^{\frac{24}{11}}, X^{\frac{3}{11}} \ll M \ll X^{\frac{4}{11}}, MH^8 \ll X^{\frac{25}{11}}, X^{\frac{63}{22}} \ll M^6H^5;$
- (4) $MH \ll X^{\frac{146}{209}}, X^{\frac{21}{121}} \ll H, M^{12}H \ll X^{\frac{69}{11}}, X^{\frac{147}{319}} \ll M, M^{-1}H^{19} \ll X^{\frac{41}{11}}, X^{\frac{84}{11}} \ll M^{12}H^{11};$
- (5) $M^2H \ll X^{\frac{38}{33}}, M^{58}H^9 \ll X^{\frac{323}{11}}, X^{\frac{14}{33}} \ll M, MH^5 \ll X^{\frac{15}{11}}, X^{\frac{315}{22}} \ll M^{29}H^{10};$
- (6) $X^{\frac{7}{11}} \ll MH \ll X^{\frac{8}{11}}, X^{\frac{21}{110}} \ll H, M^6H \ll X^{\frac{22}{11}}, X^{\frac{105}{242}} \ll M, M^{-1}H^7 \ll X^{\frac{14}{11}}, X^{\frac{42}{11}} \ll M^6H^5;$
- (7) $MH \ll X^{\frac{137}{242}}, M^{35}H^{23} \ll X^{\frac{196}{11}}, X^{\frac{105}{121}} \ll M^2H, M^2H^{13} \ll X^3, X^{\frac{378}{11}} \ll M^{70}H^{59};$
- (8) $MH \ll X^{\frac{80}{143}}, M^{41}H^{27} \ll X^{\frac{461}{22}}, X^{\frac{126}{143}} \ll M^2H, M^2H^{15} \ll X^{\frac{35}{11}}, X^{\frac{441}{11}} \ll M^{82}H^{69};$

$$(9) M^2 H \ll X^{\frac{137}{121}}, M^{70} H^{11} \ll X^{\frac{392}{11}}, X^{\frac{105}{242}} \ll M, MH^6 \ll X^{\frac{3}{2}}, X^{\frac{189}{11}} \ll M^{35} H^{12}.$$

Assume that for $(\log X)^{B/\varepsilon} \leq |t| \leq 2X$, $M(b+it) \ll (\log x)^{-B/\varepsilon}$ and $H(b+it) \ll (\log x)^{-B/\varepsilon}$. Then for $(\log X)^{B/\varepsilon} \leq T \leq X$, we have

$$\left(\min \left(\eta, \frac{1}{T} \right) \right)^2 \int_T^{2T} |M(b+it)H(b+it)K(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}.$$

Proof. Using the method of Lemma 3.1, we only need to show that for $T = 1/\eta = (2X)^{\frac{21}{22}-\varepsilon}$,

$$I = \int_T^{2T} |M(b+it)H(b+it)K(b+it)|^2 dt \ll (\log x)^{-10B}.$$

Assuming the condition (1). By applying the mean value estimate and Halász method to $M^3(s)$, $H^5(s)$ and $K^2(s)$, we get

$$I \ll U^2 V^2 W^2 x^{-1} F(\log x)^c$$

where

$$F = \min \{ V^{-6} (M^3 + T), V^{-6} M^3 + V^{-18} M^3 T, W^{-10} (H^5 + T), \\ W^{-10} H^5 + W^{-30} H^5 T, U^{-4} (K^2 + T), U^{-4} K^2 + U^{-12} K^2 T \}.$$

We consider 8 cases:

(a) $F \leq 2V^{-6} M^3, F \leq 2W^{-10} H^5, F \leq 2U^{-4} K^2$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-6} M^3, W^{-10} H^5, U^{-4} K^2 \} \\ &\leq U^2 V^2 W^2 (V^{-6} M^3)^{\frac{3}{10}} (W^{-10} H^5)^{\frac{1}{5}} (U^{-4} K^2)^{\frac{1}{2}} \\ &= V^{\frac{1}{5}} M^{\frac{9}{10}} H K \\ &\ll x(\log x)^{-11B}. \end{aligned}$$

(b) $F \leq 2V^{-6} M^3, F \leq 2W^{-10} H^5, F > 2U^{-4} K^2$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-6} M^3, W^{-10} H^5, U^{-4} T, U^{-12} K^2 T \} \\ &\leq U^2 V^2 W^2 (V^{-6} M^3)^{\frac{1}{3}} (W^{-10} H^5)^{\frac{1}{5}} (U^{-4} T)^{\frac{9}{20}} (U^{-12} K^2 T)^{\frac{1}{60}} \\ &= T^{\frac{7}{15}} M H K^{\frac{1}{30}} \\ &= T^{\frac{7}{15}} (M H K)^{\frac{1}{30}} \left(M^{\frac{29}{30}} H^{\frac{29}{30}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $MH \ll X^{\frac{172}{319}}$ is required.

(c) $F \leq 2V^{-6} M^3, F > 2W^{-10} H^5, F \leq 2U^{-4} K^2$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-6} M^3, W^{-10} T, W^{-30} H^5 T, U^{-4} K^2 \} \\ &\leq U^2 V^2 W^2 (V^{-6} M^3)^{\frac{1}{3}} (W^{-10} T)^{\frac{3}{20}} (W^{-30} H^5 T)^{\frac{1}{60}} (U^{-4} K^2)^{\frac{1}{2}} \\ &= T^{\frac{1}{6}} M H^{\frac{1}{12}} K \\ &= T^{\frac{1}{6}} (M H K) H^{-\frac{11}{12}} \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{21}{121}} \ll H$ is required.

(d) $F \leq 2V^{-6} M^3, F > 2W^{-10} H^5, F > 2U^{-4} K^2$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-6} M^3, W^{-10} T, W^{-30} H^5 T, U^{-4} T, U^{-12} K^2 T \} \\ &\leq U^2 V^2 W^2 (V^{-6} M^3)^{\frac{1}{3}} (W^{-10} T)^{\frac{1}{5}} (U^{-4} T)^{\frac{9}{20}} (U^{-12} K^2 T)^{\frac{1}{60}} \\ &= T^{\frac{2}{3}} M K^{\frac{1}{30}} \\ &= T^{\frac{2}{3}} (M H K)^{\frac{1}{30}} \left(M^{\frac{29}{30}} H^{-\frac{1}{30}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^{29} H^{-1} \ll X^{\frac{109}{11}}$ is required.

(e) $F > 2V^{-6} M^3, F \leq 2W^{-10} H^5, F \leq 2U^{-4} K^2$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-6} T, V^{-18} M^3 T, W^{-10} H^5, U^{-4} K^2 \} \\ &\leq U^2 V^2 W^2 (V^{-6} T)^{\frac{17}{60}} (V^{-18} M^3 T)^{\frac{1}{60}} (W^{-10} H^5)^{\frac{1}{5}} (U^{-4} K^2)^{\frac{1}{2}} \\ &= T^{\frac{3}{10}} M^{\frac{1}{20}} H K \end{aligned}$$

$$\begin{aligned}
&= T^{\frac{3}{10}} (MHK) M^{-\frac{19}{20}} \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{63}{209}} \ll M$ is required.

(f) $F > 2V^{-6}M^3, F \leq 2W^{-10}H^5, F > 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-6}T, V^{-18}M^3T, W^{-10}H^5, U^{-4}T, U^{-12}K^2T\} \\
&\leq U^2V^2W^2 (V^{-6}T)^{\frac{1}{3}} (W^{-10}H^5)^{\frac{1}{5}} (U^{-4}T)^{\frac{9}{20}} (U^{-12}K^2T)^{\frac{1}{60}} \\
&= T^{\frac{4}{5}} HK^{\frac{1}{30}} \\
&= T^{\frac{4}{5}} (MHK)^{\frac{1}{30}} \left(M^{-\frac{1}{30}} H^{\frac{29}{30}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^{-1}H^{29} \ll X^{\frac{67}{11}}$ is required.

(g) $F > 2V^{-6}M^3, F > 2W^{-10}H^5, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-6}T, V^{-18}M^3T, W^{-10}T, W^{-30}H^5T, U^{-4}K^2\} \\
&\leq U^2V^2W^2 (V^{-6}T)^{\frac{1}{3}} (W^{-10}T)^{\frac{3}{20}} (W^{-30}H^5T)^{\frac{1}{60}} (U^{-4}K^2)^{\frac{1}{2}} \\
&= T^{\frac{1}{2}} H^{\frac{1}{12}} K \\
&= T^{\frac{1}{2}} (MHK) \left(M^{-1} H^{-\frac{11}{12}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{63}{11}} \ll M^{12}H^{11}$ is required.

(h) $F > 2V^{-6}M^3, F > 2W^{-10}H^5, F > 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-6}, V^{-18}M^3, W^{-10}, W^{-30}H^5, U^{-4}, U^{-12}K^2\} T \\
&\leq U^2V^2W^2 (V^{-6})^{\frac{170}{60}} (V^{-18}M^3)^{\frac{1}{60}} (W^{-10})^{\frac{1}{5}} (U^{-4})^{\frac{1}{2}} T \\
&= TM^{\frac{1}{20}} \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M \ll X^{\frac{1283}{3509}}$ and $TM^{\frac{1}{20}} \ll X^{\frac{21}{22} + \frac{1}{20} \cdot \frac{1283}{3509}} \ll X^{\frac{68273}{70180}}$ are used. The former one can be obtained by using $MH \ll X^{\frac{172}{319}}$ and $X^{\frac{21}{121}} \ll H$.

Assuming the condition (2). By applying the mean value estimate and Halász method to $M^2(s)H(s), H^5(s)$ and $K^2(s)$, we get

$$I \ll U^2V^2W^2x^{-1}F(\log x)^c$$

where

$$\begin{aligned}
F &= \min \{V^{-4}W^{-2} (M^2H + T), V^{-4}W^{-2}M^2H + V^{-12}W^{-6}M^2HT, \\
&\quad W^{-10} (H^5 + T), W^{-10}H^5 + W^{-30}H^5T, U^{-4} (K^2 + T), U^{-4}K^2 + U^{-12}K^2T\}.
\end{aligned}$$

We consider 8 cases:

(a) $F \leq 2V^{-4}W^{-2}M^2H, F \leq 2W^{-10}H^5, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-10}H^5, U^{-4}K^2\} \\
&\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{3}{8}} (W^{-10}H^5)^{\frac{1}{8}} (U^{-4}K^2)^{\frac{1}{2}} \\
&= V^{\frac{1}{2}} M^{\frac{3}{4}} HK \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(b) $F \leq 2V^{-4}W^{-2}M^2H, F \leq 2W^{-10}H^5, F > 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-10}H^5, U^{-4}T, U^{-12}K^2T\} \\
&\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (W^{-10}H^5)^{\frac{1}{10}} (U^{-4}T)^{\frac{7}{20}} (U^{-12}K^2T)^{\frac{1}{20}} \\
&= T^{\frac{2}{5}} MHK^{\frac{1}{10}} \\
&= T^{\frac{2}{5}} (MHK)^{\frac{1}{10}} \left(M^{\frac{9}{10}} H^{\frac{9}{10}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $MH \ll X^{\frac{19}{33}}$ is required.

(c) $F \leq 2V^{-4}W^{-2}M^2H, F > 2W^{-10}H^5, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-10}T, W^{-30}H^5T, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= WMH^{\frac{1}{2}}K \\ &\ll x(\log x)^{-11B}. \end{aligned}$$

(d) $F \leq 2V^{-4}W^{-2}M^2H, F > 2W^{-10}H^5, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-10}T, W^{-30}H^5T, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (W^{-30}H^5T)^{\frac{1}{30}} (U^{-4}T)^{\frac{9}{20}} (U^{-12}K^2T)^{\frac{1}{10}} \\ &= T^{\frac{1}{2}}MH^{\frac{2}{3}}K^{\frac{1}{30}} \\ &= T^{\frac{1}{2}}(MHK)^{\frac{1}{30}} \left(M^{\frac{29}{30}}H^{\frac{19}{30}}\right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^{29}H^{19} \ll X^{\frac{323}{22}}$ is required.

(e) $F > 2V^{-4}W^{-2}M^2H, F \leq 2W^{-10}H^5, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}T, V^{-12}W^{-6}M^2HT, W^{-10}H^5, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}T)^{\frac{7}{20}} (V^{-12}W^{-6}M^2HT)^{\frac{1}{20}} (W^{-10}H^5)^{\frac{1}{10}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= T^{\frac{2}{5}}M^{\frac{1}{10}}H^{\frac{11}{20}}K \\ &= T^{\frac{2}{5}}(MHK) \left(M^{-\frac{9}{10}}H^{-\frac{9}{20}}\right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{28}{33}} \ll M^2H$ is required.

(f) $F > 2V^{-4}W^{-2}M^2H, F \leq 2W^{-10}H^5, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}T, V^{-12}W^{-6}M^2HT, W^{-10}H^5, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}T)^{\frac{7}{20}} (V^{-12}W^{-6}M^2HT)^{\frac{1}{20}} (W^{-10}H^5)^{\frac{1}{10}} (U^{-4}T)^{\frac{1}{2}} \\ &= T^{\frac{9}{10}} \left(M^{\frac{1}{10}}H^{\frac{11}{20}}\right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^2H^{11} \ll X^{\frac{31}{11}}$ is required. This can be obtained by using $M^2H^{11} \ll X^{\frac{30}{11}}$.

(g) $F > 2V^{-4}W^{-2}M^2H, F > 2W^{-10}H^5, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}T, V^{-12}W^{-6}M^2HT, W^{-10}T, W^{-30}H^5T, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}T)^{\frac{9}{20}} (V^{-12}W^{-6}M^2HT)^{\frac{1}{60}} (W^{-30}H^5T)^{\frac{1}{30}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= T^{\frac{1}{2}}M^{\frac{1}{30}}H^{\frac{11}{60}}K \\ &= T^{\frac{1}{2}}(MHK) \left(M^{-\frac{29}{30}}H^{-\frac{49}{60}}\right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{315}{11}} \ll M^{58}H^{49}$ is required.

(h) $F > 2V^{-4}W^{-2}M^2H, F > 2W^{-10}H^5, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}, V^{-12}W^{-6}M^2H, W^{-10}, W^{-30}H^5, U^{-4}, U^{-12}K^2\} T \\ &\leq U^2V^2W^2 (V^{-4}W^{-2})^{\frac{9}{20}} (V^{-12}W^{-6}M^2H)^{\frac{1}{60}} (W^{-30}H^5)^{\frac{1}{30}} (U^{-4})^{\frac{1}{2}} T \\ &= T \left(M^{\frac{1}{30}}H^{\frac{11}{60}}\right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^2H^{11} \ll X^{\frac{30}{11}}$ is required.

Assuming the condition (3). By applying the mean value estimate and Halász method to $M^3(s), H^4(s)$ and $K^2(s)$, we get

$$I \ll U^2V^2W^2x^{-1}F(\log x)^c$$

where

$$F = \min \{V^{-6}(M^3 + T), V^{-6}M^3 + V^{-18}M^3T, W^{-8}(H^4 + T),$$

$$W^{-8}H^4 + W^{-24}H^4T, U^{-4}(K^2 + T), U^{-4}K^2 + U^{-12}K^2T\}.$$

We consider 8 cases:

(a) $F \leq 2V^{-6}M^3, F \leq 2W^{-8}H^4, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-6}M^3, W^{-8}H^4, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-6}M^3)^{\frac{1}{4}} (W^{-8}H^4)^{\frac{1}{4}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= V^{\frac{1}{2}}M^{\frac{3}{4}}HK \\ &\ll x(\log x)^{-11B}. \end{aligned}$$

(b) $F \leq 2V^{-6}M^3, F \leq 2W^{-8}H^4, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-6}M^3, W^{-8}H^4, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-6}M^3)^{\frac{1}{3}} (W^{-8}H^4)^{\frac{1}{4}} (U^{-4}T)^{\frac{3}{8}} (U^{-12}K^2T)^{\frac{1}{24}} \\ &= T^{\frac{5}{12}}MHK^{\frac{1}{12}} \\ &= T^{\frac{5}{12}}(MHK)^{\frac{1}{12}} (M^{\frac{11}{12}}H^{\frac{11}{12}}) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $MH \ll X^{\frac{137}{242}}$ is required.

(c) $F \leq 2V^{-6}M^3, F > 2W^{-8}H^4, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-6}M^3, W^{-8}T, W^{-24}H^4T, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-6}M^3)^{\frac{1}{3}} (W^{-8}T)^{\frac{1}{8}} (W^{-24}H^4T)^{\frac{1}{24}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= T^{\frac{1}{6}}MH^{\frac{1}{6}}K \\ &= T^{\frac{1}{6}}(MHK)H^{-\frac{5}{6}} \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{21}{110}} \ll H$ is required.

(d) $F \leq 2V^{-6}M^3, F > 2W^{-8}H^4, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-6}M^3, W^{-8}T, W^{-24}H^4T, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-6}M^3)^{\frac{1}{3}} (W^{-8}T)^{\frac{1}{8}} (W^{-24}H^4T)^{\frac{1}{24}} (U^{-4}T)^{\frac{1}{2}} \\ &= T^{\frac{2}{3}}(MH^{\frac{1}{6}}) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^6H \ll X^{\frac{24}{11}}$ is required.

(e) $F > 2V^{-6}M^3, F \leq 2W^{-8}H^4, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-6}T, V^{-18}M^3T, W^{-8}H^4, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-6}T)^{\frac{5}{24}} (V^{-18}M^3T)^{\frac{1}{24}} (W^{-8}H^4)^{\frac{1}{4}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= T^{\frac{1}{4}}M^{\frac{1}{8}}HK \\ &= T^{\frac{1}{4}}(MHK)M^{-\frac{7}{8}} \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{3}{11}} \ll M$ is required.

(f) $F > 2V^{-6}M^3, F \leq 2W^{-8}H^4, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-6}T, V^{-18}M^3T, W^{-8}H^4, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-6}T)^{\frac{5}{24}} (V^{-18}M^3T)^{\frac{1}{24}} (W^{-8}H^4)^{\frac{1}{4}} (U^{-4}T)^{\frac{1}{2}} \\ &= T^{\frac{3}{4}}(M^{\frac{1}{8}}H) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $MH^8 \ll X^{\frac{25}{11}}$ is required.

(g) $F > 2V^{-6}M^3, F > 2W^{-8}H^4, F \leq 2U^{-4}K^2$. Then

$$U^2V^2W^2F \ll U^2V^2W^2 \min\{V^{-6}T, V^{-18}M^3T, W^{-8}T, W^{-24}H^4T, U^{-4}K^2\}$$

$$\begin{aligned}
&\leq U^2 V^2 W^2 (V^{-6} T)^{\frac{1}{3}} (W^{-8} T)^{\frac{1}{8}} (W^{-24} H^4 T)^{\frac{1}{24}} (U^{-4} K^2)^{\frac{1}{2}} \\
&= T^{\frac{1}{2}} H^{\frac{1}{6}} K \\
&= T^{\frac{1}{2}} (M H K) \left(M^{-1} H^{-\frac{5}{6}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{63}{22}} \ll M^6 H^5$ is required.

(h) $F > 2V^{-6} M^3, F > 2W^{-8} H^4, F > 2U^{-4} K^2$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-6}, V^{-18} M^3, W^{-8}, W^{-24} H^4, U^{-4}, U^{-12} K^2 \} T \\
&\leq U^2 V^2 W^2 (V^{-6})^{\frac{5}{24}} (V^{-18} M^3)^{\frac{1}{24}} (W^{-8})^{\frac{1}{4}} (U^{-4})^{\frac{1}{2}} T \\
&= T M^{\frac{1}{8}} \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M \ll X^{\frac{4}{11}}$ is required.

Assuming the condition (4). By applying the mean value estimate and Halász method to $M^2(s), H^5(s)$ and $K^3(s)$, we get

$$I \ll U^2 V^2 W^2 x^{-1} F(\log x)^c$$

where

$$\begin{aligned}
F &= \min \{ V^{-4} (M^2 + T), V^{-4} M^2 + V^{-12} M^2 T, W^{-10} (H^5 + T), \\
&\quad W^{-10} H^5 + W^{-30} H^5 T, U^{-6} (K^3 + T), U^{-6} K^3 + U^{-18} K^3 T \}.
\end{aligned}$$

We consider 8 cases:

(a) $F \leq 2V^{-4} M^2, F \leq 2W^{-10} H^5, F \leq 2U^{-6} K^3$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-10} H^5, U^{-6} K^3 \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-10} H^5)^{\frac{1}{6}} (U^{-6} K^3)^{\frac{1}{3}} \\
&= W^{\frac{1}{3}} M H^{\frac{5}{6}} K \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(b) $F \leq 2V^{-4} M^2, F \leq 2W^{-10} H^5, F > 2U^{-6} K^3$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-10} H^5, U^{-6} T, U^{-18} K^3 T \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-10} H^5)^{\frac{1}{6}} (U^{-6} T)^{\frac{17}{60}} (U^{-18} K^3 T)^{\frac{1}{60}} \\
&= T^{\frac{3}{10}} M H K^{\frac{1}{20}} \\
&= T^{\frac{3}{10}} (M H K)^{\frac{1}{20}} \left(M^{\frac{19}{20}} H^{\frac{19}{20}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M H \ll X^{\frac{146}{209}}$ is required.

(c) $F \leq 2V^{-4} M^2, F > 2W^{-10} H^5, F \leq 2U^{-6} K^3$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-10} T, W^{-30} H^5 T, U^{-6} K^3 \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-10} T)^{\frac{3}{20}} (W^{-30} H^5 T)^{\frac{1}{60}} (U^{-6} K^3)^{\frac{1}{3}} \\
&= T^{\frac{1}{6}} M H^{\frac{1}{12}} K \\
&= T^{\frac{1}{6}} (M H K) H^{-\frac{11}{12}} \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{21}{121}} \ll H$ is required.

(d) $F \leq 2V^{-4} M^2, F > 2W^{-10} H^5, F > 2U^{-6} K^3$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-10} T, W^{-30} H^5 T, U^{-6} T, U^{-18} K^3 T \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-10} T)^{\frac{3}{20}} (W^{-30} H^5 T)^{\frac{1}{60}} (U^{-6} T)^{\frac{1}{3}} \\
&= T^{\frac{1}{2}} \left(M H^{\frac{1}{12}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^{12} H \ll X^{\frac{69}{11}}$ is required.

(e) $F > 2V^{-4}M^2, F \leq 2W^{-10}H^5, F \leq 2U^{-6}K^3$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}T, V^{-12}M^2T, W^{-10}H^5, U^{-6}K^3\} \\
&\leq U^2V^2W^2 (V^{-4}T)^{\frac{9}{20}} (V^{-12}M^2T)^{\frac{1}{60}} (W^{-10}H^5)^{\frac{1}{5}} (U^{-6}K^3)^{\frac{1}{3}} \\
&= T^{\frac{7}{15}} M^{\frac{1}{30}} HK \\
&= T^{\frac{7}{15}} (MHK) M^{-\frac{29}{30}} \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{147}{319}} \ll M$ is required.

(f) $F > 2V^{-4}M^2, F \leq 2W^{-10}H^5, F > 2U^{-6}K^3$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}T, V^{-12}M^2T, W^{-10}H^5, U^{-6}T, U^{-18}K^3T\} \\
&\leq U^2V^2W^2 (V^{-4}T)^{\frac{1}{2}} (W^{-10}H^5)^{\frac{1}{5}} (U^{-6}T)^{\frac{17}{60}} (U^{-18}K^3T)^{\frac{1}{60}} \\
&= T^{\frac{4}{5}} HK^{\frac{1}{20}} \\
&= T^{\frac{4}{5}} (MHK)^{\frac{1}{20}} \left(M^{-\frac{1}{20}} H^{\frac{19}{20}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^{-1}H^{19} \ll X^{\frac{41}{11}}$ is required.

(g) $F > 2V^{-4}M^2, F > 2W^{-10}H^5, F \leq 2U^{-6}K^3$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}T, V^{-12}M^2T, W^{-10}T, W^{-30}H^5T, U^{-6}K^3\} \\
&\leq U^2V^2W^2 (V^{-4}T)^{\frac{1}{2}} (W^{-10}T)^{\frac{3}{20}} (W^{-30}H^5T)^{\frac{1}{60}} (U^{-6}K^3)^{\frac{1}{3}} \\
&= T^{\frac{2}{3}} H^{\frac{1}{12}} K \\
&= T^{\frac{2}{3}} (MHK) \left(M^{-1} H^{-\frac{11}{12}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{84}{11}} \ll M^{12}H^{11}$ is required.

(h) $F > 2V^{-4}M^2, F > 2W^{-10}H^5, F > 2U^{-6}K^3$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}, V^{-12}M^2, W^{-10}, W^{-30}H^5, U^{-6}, U^{-18}K^3\} T \\
&\leq U^2V^2W^2 (V^{-4})^{\frac{1}{2}} (W^{-10})^{\frac{1}{5}} (U^{-6})^{\frac{17}{60}} (U^{-18}K^3)^{\frac{1}{60}} T \\
&= TK^{\frac{1}{20}} \\
&= T(MHK)^{\frac{1}{20}} \left(M^{-\frac{1}{20}} H^{-\frac{1}{20}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{1}{11}} \ll MH$ is required. This can be obtained by using $X^{\frac{147}{319}} \ll M$ and $X^{\frac{21}{121}} \ll H$.

Assuming the condition (5). By applying the mean value estimate and Halász method to $M^2(s), H^5(s)$ and $K^2(s)H(s)$, we get

$$I \ll U^2V^2W^2x^{-1}F(\log x)^c$$

where

$$\begin{aligned}
F &= \min \{V^{-4} (M^2 + T), V^{-4}M^2 + V^{-12}M^2T, W^{-10} (H^5 + T), \\
&\quad W^{-10}H^5 + W^{-30}H^5T, U^{-4}W^{-2} (K^2H + T), U^{-4}W^{-2}K^2H + U^{-12}W^{-6}K^2HT\}.
\end{aligned}$$

We consider 8 cases:

(a) $F \leq 2V^{-4}M^2, F \leq 2W^{-10}H^5, F \leq 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}M^2, W^{-10}H^5, U^{-4}W^{-2}K^2H\} \\
&\leq U^2V^2W^2 (V^{-4}M^2)^{\frac{1}{2}} (U^{-4}W^{-2}K^2H)^{\frac{1}{2}} \\
&= WMH^{\frac{1}{2}} K \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(b) $F \leq 2V^{-4}M^2, F \leq 2W^{-10}H^5, F > 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}M^2, W^{-10}H^5, U^{-4}W^{-2}T, U^{-12}W^{-6}K^2HT\} \\
&\leq U^2V^2W^2 (V^{-4}M^2)^{\frac{1}{2}} (W^{-10}H^5)^{\frac{1}{10}} (U^{-4}W^{-2}T)^{\frac{7}{20}} (U^{-12}W^{-6}K^2HT)^{\frac{1}{20}}
\end{aligned}$$

$$\begin{aligned}
&= T^{\frac{2}{5}} M H^{\frac{11}{20}} K^{\frac{1}{10}} \\
&= T^{\frac{2}{5}} (M H K)^{\frac{1}{10}} \left(M^{\frac{9}{10}} H^{\frac{9}{20}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^2 H \ll X^{\frac{38}{33}}$ is required.

(c) $F \leq 2V^{-4}M^2, F > 2W^{-10}H^5, F \leq 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4}M^2, W^{-10}T, W^{-30}H^5T, U^{-4}W^{-2}K^2H \} \\
&\leq U^2 V^2 W^2 (V^{-4}M^2)^{\frac{1}{2}} (U^{-4}W^{-2}K^2H)^{\frac{1}{2}} \\
&= W M H^{\frac{1}{2}} K \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(d) $F \leq 2V^{-4}M^2, F > 2W^{-10}H^5, F > 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4}M^2, W^{-10}T, W^{-30}H^5T, U^{-4}W^{-2}T, U^{-12}W^{-6}K^2HT \} \\
&\leq U^2 V^2 W^2 (V^{-4}M^2)^{\frac{1}{2}} (W^{-30}H^5T)^{\frac{1}{30}} (U^{-4}W^{-2}T)^{\frac{9}{20}} (U^{-12}W^{-6}K^2HT)^{\frac{1}{60}} \\
&= T^{\frac{1}{2}} M H^{\frac{11}{60}} K^{\frac{1}{30}} \\
&= T^{\frac{1}{2}} (M H K)^{\frac{1}{30}} \left(M^{\frac{29}{30}} H^{\frac{9}{60}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^{58}H^9 \ll X^{\frac{323}{11}}$ is required.

(e) $F > 2V^{-4}M^2, F \leq 2W^{-10}H^5, F \leq 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4}T, V^{-12}M^2T, W^{-10}H^5, U^{-4}W^{-2}K^2H \} \\
&\leq U^2 V^2 W^2 (V^{-4}T)^{\frac{7}{20}} (V^{-12}M^2T)^{\frac{1}{20}} (W^{-10}H^5)^{\frac{1}{10}} (U^{-4}W^{-2}K^2H)^{\frac{1}{2}} \\
&= T^{\frac{2}{5}} M^{\frac{1}{10}} H K \\
&= T^{\frac{2}{5}} (M H K) M^{-\frac{9}{10}} \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{14}{33}} \ll M$ is required.

(f) $F > 2V^{-4}M^2, F \leq 2W^{-10}H^5, F > 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4}T, V^{-12}M^2T, W^{-10}H^5, U^{-4}W^{-2}T, U^{-12}W^{-6}K^2HT \} \\
&\leq U^2 V^2 W^2 (V^{-4}T)^{\frac{7}{20}} (V^{-12}M^2T)^{\frac{1}{20}} (W^{-10}H^5)^{\frac{1}{10}} (U^{-4}W^{-2}T)^{\frac{1}{2}} \\
&= T^{\frac{9}{10}} \left(M^{\frac{1}{10}} H^{\frac{1}{2}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M H^5 \ll X^{\frac{31}{22}}$ is required. This can be obtained by using $M H^5 \ll X^{\frac{15}{11}}$.

(g) $F > 2V^{-4}M^2, F > 2W^{-10}H^5, F \leq 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4}T, V^{-12}M^2T, W^{-10}T, W^{-30}H^5T, U^{-4}W^{-2}K^2H \} \\
&\leq U^2 V^2 W^2 (V^{-4}T)^{\frac{9}{20}} (V^{-12}M^2T)^{\frac{1}{60}} (W^{-30}H^5T)^{\frac{1}{30}} (U^{-4}W^{-2}K^2H)^{\frac{1}{2}} \\
&= T^{\frac{1}{2}} M^{\frac{1}{30}} H^{\frac{2}{3}} K \\
&= T^{\frac{1}{2}} (M H K) \left(M^{-\frac{29}{30}} H^{-\frac{1}{3}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{315}{22}} \ll M^{29}H^{10}$ is required.

(h) $F > 2V^{-4}M^2, F > 2W^{-10}H^5, F > 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4}, V^{-12}M^2, W^{-10}, W^{-30}H^5, U^{-4}W^{-2}, U^{-12}W^{-6}K^2H \} T \\
&\leq U^2 V^2 W^2 (V^{-4})^{\frac{9}{20}} (V^{-12}M^2)^{\frac{1}{60}} (W^{-30}H^5)^{\frac{1}{30}} (U^{-4}W^{-2})^{\frac{1}{2}} T \\
&= T \left(M^{\frac{1}{30}} H^{\frac{1}{6}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $MH^5 \ll X^{\frac{15}{11}}$ is required.

Assuming the condition (6). By applying the mean value estimate and Halász method to $M^2(s)$, $H^4(s)$ and $K^3(s)$, we get

$$I \ll U^2 V^2 W^2 x^{-1} F(\log x)^c$$

where

$$F = \min \{ V^{-4} (M^2 + T), V^{-4} M^2 + V^{-12} M^2 T, W^{-8} (H^4 + T), \\ W^{-8} H^4 + W^{-24} H^4 T, U^{-6} (K^3 + T), U^{-6} K^3 + U^{-18} K^3 T \}.$$

We consider 8 cases:

(a) $F \leq 2V^{-4} M^2, F \leq 2W^{-8} H^4, F \leq 2U^{-6} K^3$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-8} H^4, U^{-6} K^3 \} \\ &\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-8} H^4)^{\frac{1}{6}} (U^{-6} K^3)^{\frac{1}{3}} \\ &= W^{\frac{2}{3}} M H^{\frac{2}{3}} K \\ &\ll x(\log x)^{-11B}. \end{aligned}$$

(b) $F \leq 2V^{-4} M^2, F \leq 2W^{-8} H^4, F > 2U^{-6} K^3$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-8} H^4, U^{-6} T, U^{-18} K^3 T \} \\ &\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-8} H^4)^{\frac{1}{4}} (U^{-6} T)^{\frac{5}{24}} (U^{-18} K^3 T)^{\frac{1}{24}} \\ &= T^{\frac{1}{4}} M H K^{\frac{1}{8}} \\ &= T^{\frac{1}{4}} (M H K)^{\frac{1}{8}} \left(M^{\frac{7}{8}} H^{\frac{7}{8}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $MH \ll X^{\frac{8}{11}}$ is required.

(c) $F \leq 2V^{-4} M^2, F > 2W^{-8} H^4, F \leq 2U^{-6} K^3$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-8} T, W^{-24} H^4 T, U^{-6} K^3 \} \\ &\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-8} T)^{\frac{1}{8}} (W^{-24} H^4 T)^{\frac{1}{24}} (U^{-6} K^3)^{\frac{1}{3}} \\ &= T^{\frac{1}{6}} M H^{\frac{1}{6}} K \\ &= T^{\frac{1}{6}} (M H K) H^{-\frac{5}{6}} \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{21}{110}} \ll H$ is required.

(d) $F \leq 2V^{-4} M^2, F > 2W^{-8} H^4, F > 2U^{-6} K^3$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-8} T, W^{-24} H^4 T, U^{-6} T, U^{-18} K^3 T \} \\ &\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-8} T)^{\frac{1}{8}} (W^{-24} H^4 T)^{\frac{1}{24}} (U^{-6} T)^{\frac{1}{3}} \\ &= T^{\frac{1}{2}} \left(M H^{\frac{1}{6}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^6 H \ll X^{\frac{69}{22}}$ is required.

(e) $F > 2V^{-4} M^2, F \leq 2W^{-8} H^4, F \leq 2U^{-6} K^3$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} T, V^{-12} M^2 T, W^{-8} H^4, U^{-6} K^3 \} \\ &\leq U^2 V^2 W^2 (V^{-4} T)^{\frac{3}{8}} (V^{-12} M^2 T)^{\frac{1}{24}} (W^{-8} H^4)^{\frac{1}{4}} (U^{-6} K^3)^{\frac{1}{3}} \\ &= T^{\frac{5}{12}} M^{\frac{1}{12}} H K \\ &= T^{\frac{5}{12}} (M H K) M^{-\frac{11}{12}} \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{105}{242}} \ll M$ is required.

(f) $F > 2V^{-4} M^2, F \leq 2W^{-8} H^4, F > 2U^{-6} K^3$. Then

$$\begin{aligned} U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} T, V^{-12} M^2 T, W^{-8} H^4, U^{-6} T, U^{-18} K^3 T \} \\ &\leq U^2 V^2 W^2 (V^{-4} T)^{\frac{1}{2}} (W^{-8} H^4)^{\frac{1}{4}} (U^{-6} T)^{\frac{5}{24}} (U^{-18} K^3 T)^{\frac{1}{24}} \\ &= T^{\frac{3}{4}} H K^{\frac{1}{8}} \end{aligned}$$

$$\begin{aligned}
&= T^{\frac{3}{4}} (MHK)^{\frac{1}{8}} \left(M^{-\frac{1}{8}} H^{\frac{7}{8}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^{-1}H^7 \ll X^{\frac{14}{11}}$ is required.

(g) $F > 2V^{-4}M^2, F > 2W^{-8}H^4, F \leq 2U^{-6}K^3$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}T, V^{-12}M^2T, W^{-8}T, W^{-24}H^4T, U^{-6}K^3\} \\
&\leq U^2V^2W^2 (V^{-4}T)^{\frac{1}{2}} (W^{-8}T)^{\frac{1}{8}} (W^{-24}H^4T)^{\frac{1}{24}} (U^{-6}K^3)^{\frac{1}{3}} \\
&= T^{\frac{2}{3}} H^{\frac{1}{6}} K \\
&= T^{\frac{2}{3}} (MHK) \left(M^{-1}H^{-\frac{5}{6}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{42}{11}} \ll M^6H^5$ is required.

(h) $F > 2V^{-4}M^2, F > 2W^{-8}H^4, F > 2U^{-6}K^3$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}, V^{-12}M^2, W^{-8}, W^{-24}H^4, U^{-6}, U^{-18}K^3\} T \\
&\leq U^2V^2W^2 (V^{-4})^{\frac{1}{2}} (W^{-8})^{\frac{1}{4}} (U^{-6})^{\frac{5}{24}} (U^{-18}K^3)^{\frac{1}{24}} T \\
&= TK^{\frac{1}{8}} \\
&= T(MHK)^{\frac{1}{8}} \left(M^{-\frac{1}{8}} H^{-\frac{1}{8}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{7}{11}} \ll MH$ is required.

Assuming the condition (7). By applying the mean value estimate and Halász method to $M^2(s)H(s), H^6(s)$ and $K^2(s)$, we get

$$I \ll U^2V^2W^2x^{-1}F(\log x)^c$$

where

$$\begin{aligned}
F &= \min \{V^{-4}W^{-2}(M^2H + T), V^{-4}W^{-2}M^2H + V^{-12}W^{-6}M^2HT, \\
&\quad W^{-12}(H^6 + T), W^{-12}H^6 + W^{-36}H^6T, U^{-4}(K^2 + T), U^{-4}K^2 + U^{-12}K^2T\}.
\end{aligned}$$

We consider 8 cases:

(a) $F \leq 2V^{-4}W^{-2}M^2H, F \leq 2W^{-12}H^6, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-12}H^6, U^{-4}K^2\} \\
&\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (U^{-4}K^2)^{\frac{1}{2}} \\
&= WMH^{\frac{1}{2}}K \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(b) $F \leq 2V^{-4}W^{-2}M^2H, F \leq 2W^{-12}H^6, F > 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-12}H^6, U^{-4}T, U^{-12}K^2T\} \\
&\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (W^{-12}H^6)^{\frac{1}{12}} (U^{-4}T)^{\frac{3}{8}} (U^{-12}K^2T)^{\frac{1}{24}} \\
&= T^{\frac{5}{12}} MHK^{\frac{1}{12}} \\
&= T^{\frac{5}{12}} (MHK)^{\frac{1}{12}} \left(M^{\frac{11}{12}} H^{\frac{11}{12}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $MH \ll X^{\frac{137}{242}}$ is required.

(c) $F \leq 2V^{-4}W^{-2}M^2H, F > 2W^{-12}H^6, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned}
U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-12}T, W^{-36}H^6T, U^{-4}K^2\} \\
&\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (U^{-4}K^2)^{\frac{1}{2}} \\
&= WMH^{\frac{1}{2}}K \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(d) $F \leq 2V^{-4}W^{-2}M^2H, F > 2W^{-12}H^6, F > 2U^{-4}K^2$. Then

$$U^2V^2W^2F \ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-12}T, W^{-36}H^6T, U^{-4}T, U^{-12}K^2T\}$$

$$\begin{aligned}
&\leq U^2 V^2 W^2 (V^{-4} W^{-2} M^2 H)^{\frac{1}{2}} (W^{-36} H^6 T)^{\frac{1}{36}} (U^{-4} T)^{\frac{11}{24}} (U^{-12} K^2 T)^{\frac{1}{72}} \\
&= T^{\frac{1}{2}} M H^{\frac{2}{3}} K^{\frac{1}{36}} \\
&= T^{\frac{1}{2}} (M H K)^{\frac{1}{36}} \left(M^{\frac{35}{36}} H^{\frac{23}{36}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^{35} H^{23} \ll X^{\frac{190}{11}}$ is required.

(e) $F > 2V^{-4} W^{-2} M^2 H, F \leq 2W^{-12} H^6, F \leq 2U^{-4} K^2$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} W^{-2} T, V^{-12} W^{-6} M^2 H T, W^{-12} H^6, U^{-4} K^2 \} \\
&\leq U^2 V^2 W^2 (V^{-4} W^{-2} T)^{\frac{3}{8}} (V^{-12} W^{-6} M^2 H T)^{\frac{1}{24}} (W^{-12} H^6)^{\frac{1}{12}} (U^{-4} K^2)^{\frac{1}{2}} \\
&= T^{\frac{5}{12}} M^{\frac{1}{12}} H^{\frac{13}{24}} K \\
&= T^{\frac{5}{12}} (M H K) \left(M^{-\frac{11}{12}} H^{-\frac{11}{24}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{105}{121}} \ll M^2 H$ is required.

(f) $F > 2V^{-4} W^{-2} M^2 H, F \leq 2W^{-12} H^6, F > 2U^{-4} K^2$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} W^{-2} T, V^{-12} W^{-6} M^2 H T, W^{-12} H^6, U^{-4} T, U^{-12} K^2 T \} \\
&\leq U^2 V^2 W^2 (V^{-4} W^{-2} T)^{\frac{3}{8}} (V^{-12} W^{-6} M^2 H T)^{\frac{1}{24}} (W^{-12} H^6)^{\frac{1}{12}} (U^{-4} T)^{\frac{1}{2}} \\
&= T^{\frac{11}{12}} \left(M^{\frac{1}{12}} H^{\frac{13}{24}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^2 H^{13} \ll X^3$ is required.

(g) $F > 2V^{-4} W^{-2} M^2 H, F > 2W^{-12} H^6, F \leq 2U^{-4} K^2$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} W^{-2} T, V^{-12} W^{-6} M^2 H T, W^{-12} T, W^{-36} H^6 T, U^{-4} K^2 \} \\
&\leq U^2 V^2 W^2 (V^{-4} W^{-2} T)^{\frac{11}{24}} (V^{-12} W^{-6} M^2 H T)^{\frac{1}{72}} (W^{-36} H^6 T)^{\frac{1}{36}} (U^{-4} K^2)^{\frac{1}{2}} \\
&= T^{\frac{1}{2}} M^{\frac{1}{36}} H^{\frac{13}{72}} K \\
&= T^{\frac{1}{2}} (M H K) \left(M^{-\frac{35}{36}} H^{-\frac{59}{72}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{378}{11}} \ll M^{70} H^{59}$ is required.

(h) $F > 2V^{-4} W^{-2} M^2 H, F > 2W^{-12} H^6, F > 2U^{-4} K^2$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} W^{-2}, V^{-12} W^{-6} M^2 H, W^{-12}, W^{-36} H^6, U^{-4}, U^{-12} K^2 \} T \\
&\leq U^2 V^2 W^2 (V^{-4} W^{-2})^{\frac{1}{2}} (W^{-12})^{\frac{1}{12}} (U^{-4})^{\frac{3}{8}} (U^{-12} K^2)^{\frac{1}{24}} T \\
&= T K^{\frac{1}{12}} \\
&= T (M H K)^{\frac{1}{12}} \left(M^{-\frac{1}{12}} H^{-\frac{1}{12}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{5}{11}} \ll M H$ is required. This can be obtained by using $X^{\frac{27}{55}} \ll M H^{\frac{59}{70}} \ll M H$.

Assuming the condition (8). By applying the mean value estimate and Halász method to $M^2(s)H(s), H^7(s)$ and $K^2(s)$, we get

$$I \ll U^2 V^2 W^2 x^{-1} F(\log x)^c$$

where

$$\begin{aligned}
F &= \min \{ V^{-4} W^{-2} (M^2 H + T), V^{-4} W^{-2} M^2 H + V^{-12} W^{-6} M^2 H T, \\
&\quad W^{-14} (H^7 + T), W^{-14} H^7 + W^{-42} H^7 T, U^{-4} (K^2 + T) U^{-4} K^2 + U^{-12} K^2 T \}.
\end{aligned}$$

We consider 8 cases:

(a) $F \leq 2V^{-4} W^{-2} M^2 H, F \leq 2W^{-14} H^7, F \leq 2U^{-4} K^2$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} W^{-2} M^2 H, W^{-14} H^7, U^{-4} K^2 \} \\
&\leq U^2 V^2 W^2 (V^{-4} W^{-2} M^2 H)^{\frac{1}{2}} (U^{-4} K^2)^{\frac{1}{2}} \\
&= W M H^{\frac{1}{2}} K
\end{aligned}$$

$$\ll x(\log x)^{-11B}.$$

(b) $F \leq 2V^{-4}W^{-2}M^2H, F \leq 2W^{-14}H^7, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-14}H^7, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (W^{-14}H^7)^{\frac{1}{14}} (U^{-4}T)^{\frac{11}{28}} (U^{-12}K^2T)^{\frac{1}{28}} \\ &= T^{\frac{3}{7}} M H K^{\frac{1}{14}} \\ &= T^{\frac{3}{7}} (M H K)^{\frac{1}{14}} \left(M^{\frac{13}{14}} H^{\frac{13}{14}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $MH \ll X^{\frac{80}{143}}$ is required.

(c) $F \leq 2V^{-4}W^{-2}M^2H, F > 2W^{-14}H^7, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-14}T, W^{-42}H^7T, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= W M H^{\frac{1}{2}} K \\ &\ll x(\log x)^{-11B}. \end{aligned}$$

(d) $F \leq 2V^{-4}W^{-2}M^2H, F > 2W^{-14}H^7, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}M^2H, W^{-14}T, W^{-42}H^7T, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}M^2H)^{\frac{1}{2}} (W^{-42}H^7T)^{\frac{1}{42}} (U^{-4}T)^{\frac{13}{28}} (U^{-12}K^2T)^{\frac{1}{84}} \\ &= T^{\frac{1}{2}} M H^{\frac{2}{3}} K^{\frac{1}{42}} \\ &= T^{\frac{1}{2}} (M H K)^{\frac{1}{42}} \left(M^{\frac{41}{42}} H^{\frac{9}{14}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^{41}H^{27} \ll X^{\frac{461}{22}}$ is required.

(e) $F > 2V^{-4}W^{-2}M^2H, F \leq 2W^{-14}H^7, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}T, V^{-12}W^{-6}M^2HT, W^{-14}H^7, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}T)^{\frac{11}{28}} (V^{-12}W^{-6}M^2HT)^{\frac{1}{28}} (W^{-14}H^7)^{\frac{1}{14}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= T^{\frac{3}{7}} M^{\frac{1}{14}} H^{\frac{15}{28}} K \\ &= T^{\frac{3}{7}} (M H K) \left(M^{-\frac{13}{14}} H^{-\frac{13}{28}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{126}{143}} \ll M^2H$ is required.

(f) $F > 2V^{-4}W^{-2}M^2H, F \leq 2W^{-14}H^7, F > 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}T, V^{-12}W^{-6}M^2HT, W^{-14}H^7, U^{-4}T, U^{-12}K^2T\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}T)^{\frac{11}{28}} (V^{-12}W^{-6}M^2HT)^{\frac{1}{28}} (W^{-14}H^7)^{\frac{1}{14}} (U^{-4}T)^{\frac{1}{2}} \\ &= T^{\frac{13}{14}} \left(M^{\frac{1}{14}} H^{\frac{15}{28}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M^2H^{15} \ll X^{\frac{35}{11}}$ is required.

(g) $F > 2V^{-4}W^{-2}M^2H, F > 2W^{-14}H^7, F \leq 2U^{-4}K^2$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min \{V^{-4}W^{-2}T, V^{-12}W^{-6}M^2HT, W^{-14}T, W^{-42}H^7T, U^{-4}K^2\} \\ &\leq U^2V^2W^2 (V^{-4}W^{-2}T)^{\frac{13}{88}} (V^{-12}W^{-6}M^2HT)^{\frac{1}{84}} (W^{-42}H^7T)^{\frac{1}{42}} (U^{-4}K^2)^{\frac{1}{2}} \\ &= T^{\frac{1}{2}} M^{\frac{1}{42}} H^{\frac{5}{28}} K \\ &= T^{\frac{1}{2}} (M H K) \left(M^{-\frac{41}{42}} H^{-\frac{23}{28}} \right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{441}{11}} \ll M^{82}H^{69}$ is required.

(h) $F > 2V^{-4}W^{-2}M^2H, F > 2W^{-14}H^7, F > 2U^{-4}K^2$. Then

$$U^2V^2W^2F \ll U^2V^2W^2 \min \{V^{-4}W^{-2}, V^{-12}W^{-6}M^2H, W^{-14}, W^{-42}H^7, U^{-4}, U^{-12}K^2\} T$$

$$\begin{aligned}
&\leq U^2 V^2 W^2 (V^{-4} W^{-2})^{\frac{1}{2}} (W^{-14})^{\frac{1}{14}} (U^{-4})^{\frac{11}{28}} (U^{-12} K^2)^{\frac{1}{28}} T \\
&= T K^{\frac{1}{14}} \\
&= T (M H K)^{\frac{1}{14}} \left(M^{-\frac{1}{14}} H^{-\frac{1}{14}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{4}{11}} \ll M H$ is required. This can be obtained by using $X^{\frac{63}{143}} \ll M H^{\frac{1}{2}} \ll M H$.

Assuming the condition (9). By applying the mean value estimate and Halász method to $M^2(s)$, $H^6(s)$ and $K^2(s)H(s)$, we get

$$I \ll U^2 V^2 W^2 x^{-1} F(\log x)^c$$

where

$$\begin{aligned}
F = \min \{ &V^{-4} (M^2 + T), V^{-4} M^2 + V^{-12} M^2 T, W^{-12} (H^6 + T), W^{-12} H^6 + W^{-36} H^6 T, \\
&U^{-4} W^{-2} (K^2 H + T), U^{-4} W^{-2} K^2 H + U^{-12} W^{-6} K^2 H T \}.
\end{aligned}$$

We consider 8 cases:

(a) $F \leq 2V^{-4} M^2$, $F \leq 2W^{-12} H^6$, $F \leq 2U^{-4} W^{-2} K^2 H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-12} H^6, U^{-4} W^{-2} K^2 H \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (U^{-4} W^{-2} K^2 H)^{\frac{1}{2}} \\
&= W M H^{\frac{1}{2}} K \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(b) $F \leq 2V^{-4} M^2$, $F \leq 2W^{-12} H^6$, $F > 2U^{-4} W^{-2} K^2 H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-12} H^6, U^{-4} W^{-2} T, U^{-12} W^{-6} K^2 H T \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-12} H^6)^{\frac{1}{12}} (U^{-4} W^{-2} T)^{\frac{3}{8}} (U^{-12} W^{-6} K^2 H T)^{\frac{1}{24}} \\
&= T^{\frac{5}{12}} M H^{\frac{13}{24}} K^{\frac{1}{12}} \\
&= T^{\frac{5}{12}} (M H K)^{\frac{1}{12}} \left(M^{\frac{11}{12}} H^{\frac{11}{24}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^2 H \ll X^{\frac{137}{121}}$ is required.

(c) $F \leq 2V^{-4} M^2$, $F > 2W^{-12} H^6$, $F \leq 2U^{-4} W^{-2} K^2 H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-12} T, W^{-36} H^6 T, U^{-4} W^{-2} K^2 H \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (U^{-4} W^{-2} K^2 H)^{\frac{1}{2}} \\
&= W M H^{\frac{1}{2}} K \\
&\ll x(\log x)^{-11B}.
\end{aligned}$$

(d) $F \leq 2V^{-4} M^2$, $F > 2W^{-12} H^6$, $F > 2U^{-4} W^{-2} K^2 H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} M^2, W^{-12} T, W^{-36} H^6 T, U^{-4} W^{-2} T, U^{-12} W^{-6} K^2 H T \} \\
&\leq U^2 V^2 W^2 (V^{-4} M^2)^{\frac{1}{2}} (W^{-36} H^6 T)^{\frac{1}{36}} (U^{-4} W^{-2} T)^{\frac{11}{24}} (U^{-12} W^{-6} K^2 H T)^{\frac{1}{72}} \\
&= T^{\frac{1}{2}} M H^{\frac{13}{72}} K^{\frac{1}{36}} \\
&= T^{\frac{1}{2}} (M H K)^{\frac{1}{36}} \left(M^{\frac{35}{36}} H^{\frac{11}{72}} \right) \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $M^{70} H^{11} \ll X^{\frac{392}{11}}$ is required.

(e) $F > 2V^{-4} M^2$, $F \leq 2W^{-12} H^6$, $F \leq 2U^{-4} W^{-2} K^2 H$. Then

$$\begin{aligned}
U^2 V^2 W^2 F &\ll U^2 V^2 W^2 \min \{ V^{-4} T, V^{-12} M^2 T, W^{-12} H^6, U^{-4} W^{-2} K^2 H \} \\
&\leq U^2 V^2 W^2 (V^{-4} T)^{\frac{3}{8}} (V^{-12} M^2 T)^{\frac{1}{24}} (W^{-12} H^6)^{\frac{1}{12}} (U^{-4} W^{-2} K^2 H)^{\frac{1}{2}} \\
&= T^{\frac{5}{12}} M^{\frac{1}{12}} H K \\
&= T^{\frac{5}{12}} (M H K) M^{-\frac{11}{12}} \\
&\ll x^{1-\varepsilon^2},
\end{aligned}$$

where $X^{\frac{105}{242}} \ll M$ is required.

(f) $F > 2V^{-4}M^2, F \leq 2W^{-12}H^6, F > 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-4}T, V^{-12}M^2T, W^{-12}H^6, U^{-4}W^{-2}T, U^{-12}W^{-6}K^2HT\} \\ &\leq U^2V^2W^2 (V^{-4}T)^{\frac{3}{8}} (V^{-12}M^2T)^{\frac{1}{24}} (W^{-12}H^6)^{\frac{1}{12}} (U^{-4}W^{-2}T)^{\frac{1}{2}} \\ &= T^{\frac{11}{12}} \left(M^{\frac{1}{12}} H^{\frac{1}{2}}\right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $MH^6 \ll X^{\frac{3}{2}}$ is required.

(g) $F > 2V^{-4}M^2, F > 2W^{-12}H^6, F \leq 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-4}T, V^{-12}M^2T, W^{-12}T, W^{-36}H^6T, U^{-4}W^{-2}K^2H\} \\ &\leq U^2V^2W^2 (V^{-4}T)^{\frac{11}{24}} (V^{-12}M^2T)^{\frac{1}{72}} (W^{-36}H^6T)^{\frac{1}{36}} (U^{-4}W^{-2}K^2H)^{\frac{1}{2}} \\ &= T^{\frac{1}{2}} M^{\frac{1}{36}} H^{\frac{2}{3}} K \\ &= T^{\frac{1}{2}} (MHK) \left(M^{-\frac{35}{36}} H^{-\frac{1}{3}}\right) \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $X^{\frac{189}{11}} \ll M^{35}H^{12}$ is required.

(h) $F > 2V^{-4}M^2, F > 2W^{-12}H^6, F > 2U^{-4}W^{-2}K^2H$. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2 \min\{V^{-4}, V^{-12}M^2, W^{-12}, W^{-36}H^6, U^{-4}W^{-2}, U^{-12}W^{-6}K^2H\} T \\ &\leq U^2V^2W^2 (V^{-4})^{\frac{3}{8}} (V^{-12}M^2)^{\frac{1}{24}} (W^{-12})^{\frac{1}{12}} (U^{-4}W^{-2})^{\frac{1}{2}} T \\ &= TM^{\frac{1}{12}} \\ &\ll x^{1-\varepsilon^2}, \end{aligned}$$

where $M \ll X^{\frac{6}{11}}$ is required. This can be obtained by using $M \ll MH^{\frac{11}{70}} \ll X^{\frac{28}{55}}$.

Finally, by combining all the cases above, Lemma 3.2 is proved. $\square$

Lemma 3.3. Suppose that $NQRK = X$, and N, Q and R satisfy one of the following 11 conditions:

- (1) $X^{\frac{339}{836}} \ll NQ \ll X^{\frac{6999}{17138}}, (NQ)^{-\frac{82}{69}} X^{\frac{147}{253}} \ll R \ll Q \ll (NQ)^{-\frac{35}{23}} X^{\frac{196}{253}};$
- (2) $X^{\frac{6999}{17138}} \ll NQ \ll X^{\frac{59}{143}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{35}{23}} X^{\frac{196}{253}};$
- (3) $X^{\frac{59}{143}} \ll NQ \ll X^{\frac{239}{572}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-1} X^{\frac{80}{143}};$
- (4) $X^{\frac{239}{572}} \ll NQ \ll X^{\frac{47}{110}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{41}{27}} X^{\frac{461}{594}};$
- (5) $X^{\frac{379}{836}} \ll NQ \ll X^{\frac{5}{11}}, (NQ)^{-\frac{35}{12}} X^{\frac{63}{44}} \ll R \ll Q \ll (NQ)^{-\frac{1}{5}} X^{\frac{3}{11}};$
- (6) $X^{\frac{31}{66}} \ll NQ \ll X^{\frac{255}{539}}, X^{\frac{2}{11}} \ll R \ll Q \ll (NQ)^{\frac{1}{7}} X^{\frac{2}{11}};$
- (7) $X^{\frac{255}{539}} \ll NQ \ll X^{\frac{21}{44}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{\frac{1}{7}} X^{\frac{2}{11}};$
- (8) $X^{\frac{21}{44}} \ll NQ \ll X^{\frac{58}{121}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-1} X^{\frac{8}{11}};$
- (9) $X^{\frac{243}{506}} \ll NQ \ll X^{\frac{1281}{2662}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{1}{6}} X^{\frac{1}{4}};$
- (10) $X^{\frac{1281}{2662}} \ll NQ \ll X^{\frac{85}{176}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-2} X^{\frac{137}{121}};$
- (11) $X^{\frac{85}{176}} \ll NQ \ll X^{\frac{139}{286}}, X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{70}{11}} X^{\frac{392}{121}}.$

Assume further that for $(\log X)^{B/\varepsilon} \leq |t| \leq 2X, N(b+it)Q(b+it) \ll (\log x)^{-B/\varepsilon}$ and $R(b+it) \ll (\log x)^{-B/\varepsilon}$. Then for $(\log X)^{B/\varepsilon} \leq T \leq X$, we have

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 \int_T^{2T} |N(b+it)Q(b+it)R(b+it)K(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}.$$

Proof. Let $M(s) = N(s)Q(s)$ and $H(s) = R(s)$. Then by Lemma 3.2, Lemma 3.3 is proved. $\square$

Lemma 3.4. Suppose that $NQRK = X$, and N, Q and R satisfy $X^{\frac{20}{209}} \ll R \ll Q$ and one of the following 3 conditions:

- (1) $N(QR) \ll X^{\frac{172}{319}}, X^{\frac{21}{121}} \ll (QR), N^{29}(QR)^{-1} \ll X^{\frac{109}{11}}, X^{\frac{63}{209}} \ll N, N^{-1}(QR)^{29} \ll X^{\frac{67}{11}}, X^{\frac{63}{11}} \ll N^{12}(QR)^{11};$
- (2) $N(QR) \ll X^{\frac{137}{242}}, X^{\frac{21}{110}} \ll (QR), N^6(QR) \ll X^{\frac{24}{11}}, X^{\frac{3}{11}} \ll N \ll X^{\frac{4}{11}}, N(QR)^8 \ll X^{\frac{25}{11}}, X^{\frac{63}{22}} \ll N^6(QR)^5;$
- (3) $N(QR) \ll X^{\frac{146}{209}}, X^{\frac{21}{121}} \ll (QR), N^{12}(QR) \ll X^{\frac{69}{11}}, X^{\frac{147}{319}} \ll N, N^{-1}(QR)^{19} \ll X^{\frac{41}{11}}, X^{\frac{84}{11}} \ll N^{12}(QR)^{11}.$

Assume further that for $(\log X)^{B/\varepsilon} \leq |t| \leq 2X, Q(b+it)R(b+it) \ll (\log x)^{-B/\varepsilon}$ and $N(b+it) \ll (\log x)^{-B/\varepsilon}$. Then for $(\log X)^{B/\varepsilon} \leq T \leq X$, we have

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 \int_T^{2T} |N(b+it)Q(b+it)R(b+it)K(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}.$$

Proof. Let $M(s) = N(s)$ and $H(s) = Q(s)R(s)$ and by Lemma 3.2 (Cases 1,3,4), Lemma 3.4 is proved. $\square$

4. ARITHMETIC INFORMATION II

In this section we are looking for more Type-I information. In [26], Jia used a mean value bound of Deshouillers and Iwaniec [8], which makes an approximation to the sixth-power moment of the Riemann zeta-function. Now we shall use another mean value bound of Watt [43], which is stronger than that of Deshouillers and Iwaniec when the length of interval is $n^{\frac{1}{22}+\varepsilon}$.

Lemma 4.1. *Let $T \geq 1$, then*

$$\int_T^{2T} \left| L\left(\frac{1}{2} + it\right) \right|^4 \left| N\left(\frac{1}{2} + it\right) \right|^2 dt \ll (T + N^2 T^{\frac{1}{2}} + N L^2 T^{-2}) T^{\varepsilon^2}.$$

Proof. For $c_1 L \leq T^{\frac{1}{2}}$ and $N \leq T$, by the main theorem proved in [43] we have

$$\int_T^{2T} \left| L\left(\frac{1}{2} + it\right) \right|^4 \left| N\left(\frac{1}{2} + it\right) \right|^2 dt \ll (T + N^2 T^{\frac{1}{2}}) T^{\varepsilon^2}.$$

For $c_1 L \leq T^{\frac{1}{2}}$ and $N > T$, by the classical mean value estimate (see [26]) we have

$$\begin{aligned} \int_T^{2T} \left| L\left(\frac{1}{2} + it\right) \right|^4 \left| N\left(\frac{1}{2} + it\right) \right|^2 dt &\ll (L^2 N + T) (LN)^{\varepsilon^2} \\ &\ll (T + N^2 T^{\frac{1}{2}}) T^{\varepsilon^2}. \end{aligned}$$

When $T^{\frac{1}{2}} < c_1 L \leq 2T$, by a reflection principle based on an approximate functional equation, we can get the same bound as above. By this process we may replace L by $L_0 \sim T/L$, so that $L_0 \ll T^{\frac{1}{2}}$. For this, one can see [[8], Section 2], [[4], Lemma 2] or [[1], Lemma 5.2] for a detailed proof with $N \leq T$. For $N > T$, by the classical mean value estimate we have

$$\begin{aligned} \int_T^{2T} \left| L\left(\frac{1}{2} + it\right) \right|^4 \left| N\left(\frac{1}{2} + it\right) \right|^2 dt &\ll (L_0^2 N + T) (L_0 N)^{\varepsilon^2} + \text{Error} \\ &\ll (T + N^2 T^{\frac{1}{2}}) T^{\varepsilon^2}. \end{aligned}$$

When $2T < c_1 L$, by the method in [26] we have

$$\int_T^{2T} \left| L\left(\frac{1}{2} + it\right) \right|^4 \left| N\left(\frac{1}{2} + it\right) \right|^2 dt \ll N L^2 T^{-2}.$$

(Actually we have a better bound $L^2 T^{-4}(N + T)$ in this case, but this won't bring any improvement to the next lemma.)

Finally, by combining all the cases above, Lemma 4.1 is proved. $\square$

Lemma 4.2. *Suppose that $MHL = X$, and M and H satisfy one of the following 2 conditions:*

- (1) $M^2 H \ll X^{\frac{23}{22}}$, $M^4 H^6 \ll X^{\frac{67}{22}}$, $H^4 \ll X^{\frac{25}{22}}$;
- (2) $M \ll X^{\frac{23}{44}}$, $H \ll X^{\frac{21}{176}}$.

Then for $\sqrt{L} \leq T \leq X$, we have

$$\left(\min\left(\eta, \frac{1}{T}\right) \right)^2 \int_T^{2T} |M(b+it)H(b+it)L(b+it)|^2 dt \ll \eta^2 x^{-2\varepsilon^2}.$$

Proof. We can prove this by using Lemma 4.1 and the methods in [[26], Lemmas 3,4]. For condition (1) we apply Lemma 4.1 to L and N , while for condition (2) we apply Lemma 4.1 to L and N^2 . One can see [26] for an explanation. $\square$

Lemma 4.3. *Suppose that $NQRL = X$, and N , Q and R satisfy the following condition:*

$$X^{\frac{85}{198}} \ll NQ \ll X^{\frac{255}{539}}, \quad X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{1}{5}} X^{\frac{3}{11}}.$$

Assume further that for $\sqrt{L} \leq |t| \leq 2X$, $N(b+it)Q(b+it) \ll (\log x)^{-B/\varepsilon}$ and $R(b+it) \ll (\log x)^{-B/\varepsilon}$. Then for $\sqrt{L} \leq T \leq X$, we have

$$\left(\min\left(\eta, \frac{1}{T}\right) \right)^2 \int_T^{2T} |N(b+it)Q(b+it)R(b+it)L(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}.$$

Proof. Let $M(s) = N(s)Q(s)$ and $H(s) = R(s)$. If $R \gg \max\left((NQ)^{-2} X^{\frac{23}{22}}, X^{\frac{21}{176}}\right)$, we apply Lemma 3.2 (Case 5). Otherwise we apply Lemma 4.2. Then Lemma 4.3 is proved. $\square$

5. ARITHMETIC INFORMATION III

In this section we provide the “extra” arithmetic information proved by Harman [[14], Chapter 9]. Some parts of it bring new asymptotic regions. In the next section we shall show that it is useful when combining with the variable role-reversal.

Lemma 5.1. ([14], Lemma 9.3)]. Let $T_0 = \exp((\log x)^{1/3})$ and $T = X^\varepsilon/\eta$. Suppose that

$$\int_{T_0}^T \left| M\left(\frac{1}{2} + it\right) \right|^2 dt \ll X(\log x)^{-K_0},$$

then

$$\sum_{k \in \mathcal{A}} a_m = \frac{\eta}{\delta} (1 + o(1)) \sum_{k \in \mathcal{B}^2} a_m$$

for all $x \in [X, 2X]$ except for $O(X(\log X)^{-K_0})$ values.

Lemma 5.2. ([14], Lemma 9.4)]. Let $T_0 = \exp((\log x)^{1/3})$ and $T = X^\varepsilon/\eta$. Suppose that $MHK = X$, $M = X^{\alpha_1}$, $H = X^{\alpha_2}$ and $0 < \alpha_2 \leq \alpha_1 < \frac{1}{2}$. Suppose that M , H and K satisfy one of the following 3 conditions:

- (1) $\alpha_2 > \frac{1}{11}$, $2\alpha_2 - \frac{1}{11} < \alpha_1 \leq \frac{2}{11}$, $M(s)$ prime-factored;
- (2) $|4\alpha_1 + 3\alpha_2 - 2| < \frac{1}{11}$, $\frac{1}{11} \leq \alpha_2 < \frac{2}{11}$, $H(s)$ prime-factored;
- (3) $|4\alpha_1 + \alpha_2 - 2| < \frac{1}{11}$, $\frac{1}{11} \leq \alpha_2 < \frac{2}{11}$, $H(s)$ prime-factored.

Then we have

$$\int_{T_0}^T \left| M\left(\frac{1}{2} + it\right) |H\left(\frac{1}{2} + it\right) |K\left(\frac{1}{2} + it\right) \right|^2 dt \ll X(\log x)^{-K_0}.$$

Lemma 5.3. Let $T_0 = \exp((\log x)^{1/3})$ and $T = X^\varepsilon/\eta$. Suppose that $NQRK = X$, $R(s)$ is prime-factored, and N , Q and R satisfy one of the following 6 conditions:

- (1) $X^{\frac{339}{836}} \ll NQ \ll X^{\frac{6999}{17138}}$, $X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{82}{69}} X^{\frac{147}{253}}$;
- (2) $X^{\frac{47}{110}} \ll NQ \ll X^{\frac{377}{836}}$, $X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{4}{3}} X^{\frac{23}{33}}$;
- (3) $X^{\frac{379}{836}} \ll NQ \ll X^{\frac{5}{11}}$, $X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{35}{12}} X^{\frac{63}{44}}$;
- (4) $X^{\frac{5}{11}} \ll NQ \ll X^{\frac{255}{539}}$, $X^{\frac{20}{209}} \ll R \ll Q \ll X^{\frac{2}{11}}$;
- (5) $X^{\frac{58}{121}} \ll NQ \ll X^{\frac{243}{506}}$, $X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-4} X^{\frac{23}{11}}$;
- (6) $X^{\frac{139}{286}} \ll NQ \ll X^{\frac{417}{836}}$, $X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-4} X^{\frac{23}{11}}$.

Then we have

$$\int_{T_0}^T \left| N\left(\frac{1}{2} + it\right) Q\left(\frac{1}{2} + it\right) R\left(\frac{1}{2} + it\right) K\left(\frac{1}{2} + it\right) \right|^2 dt \ll X(\log x)^{-K_0}.$$

Proof. Let $M(s) = N(s)Q(s)$ and $H(s) = R(s)$. Then by Lemma 5.2, Lemma 5.3 is proved. $\square$

6. SIEVE ASYMPTOTIC FORMULAS

In this section we transform the arithmetic given in Sections 3–5 into asymptotic formulas of the form (3). The first two lemmas deal with the case $z = X^\delta$.

Lemma 6.1. Suppose that $MHL = X$, and M and H satisfy one of the 2 conditions in Lemma 4.2. Then for real numbers $x \in [X, 2X]$, except for $O(X(\log X)^{-B})$ values, we have

$$\sum_{\substack{m \sim M \\ h \sim H}} a_m b_h \left(\sum_{x - \eta x \leq mhl < x} 1 - \frac{\eta x}{mh} \right) \ll \eta x (\log x)^{-B}.$$

Proof. The statement is trivial when $MH \leq X^{\frac{1}{22}}$, hence we can suppose that $MH > X^{\frac{1}{22}}$. Let $F(s) = M(s)H(s)L(s)$. By Perron's formula, we have

$$\sum_{\substack{x - \eta x \leq mhl < x \\ m \sim M \\ h \sim H}} a_m b_h = \frac{1}{2\pi i} \int_{b-iX}^{b+iX} F(s) \frac{1 - (1 - \eta)^s}{s} x^s ds + O(x^\varepsilon). \quad (5)$$

When $s = b + it$ with $|t| \leq c_1 L$, we know that

$$L(s) = \frac{(c_2 L)^{1-s} - (c_1 L)^{1-s}}{1-s} + O(L^{-1}). \quad (6)$$

Moreover, since

$$(1 - \eta)^s = \exp(s \log(1 - \eta)) = e^{-s\eta} (1 + O(|s|\eta^2)), \quad (7)$$

we have

$$\frac{1 - (1 - \eta)^s}{s} = \begin{cases} \eta + O(|s|\eta^2), & |s|\eta \ll 1, \\ O\left(\frac{1}{|s|}\right), & |s|\eta \gg 1. \end{cases} \quad (8)$$

We know that

$$\begin{aligned} \frac{1}{2\pi i} \int_{b-iX}^{b+iX} F(s) \frac{1 - (1 - \eta)^s}{s} x^s ds &= \frac{1}{2\pi i} \int_{b-i\sqrt{L}}^{b+i\sqrt{L}} F(s) \frac{1 - (1 - \eta)^s}{s} x^s ds \\ &\quad + \frac{1}{2\pi i} \int_{b+i\sqrt{L}}^{b+iX} F(s) \frac{1 - (1 - \eta)^s}{s} x^s ds \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2\pi i} \int_{b-iX}^{b-i\sqrt{L}} F(s) \frac{1-(1-\eta)^s}{s} x^s ds \\
& = I_1 + I_2 + I_3.
\end{aligned} \tag{9}$$

Since $L = \frac{X}{MH} \leq X^{\frac{21}{22}}$, we know that $|s|\eta \ll 1$ in I_1 . By (6)–(8), for I_1 we have

$$\begin{aligned}
I_1 &= \frac{1}{2\pi i} \int_{b-i\sqrt{L}}^{b+i\sqrt{L}} F(s) \frac{1-(1-\eta)^s}{s} x^s ds = \frac{\eta}{2\pi i} \int_{b-i\sqrt{L}}^{b+i\sqrt{L}} M(s)H(s) \frac{(c_2L)^{1-s} - (c_1L)^{1-s}}{1-s} x^s ds \\
&+ O\left(\eta x L^{-1} \int_{-\sqrt{L}}^{\sqrt{L}} |M(b+it)H(b+it)| dt\right) \\
&+ O\left(\eta^2 x \int_{-\sqrt{L}}^{\sqrt{L}} |M(b+it)H(b+it)| dt\right).
\end{aligned} \tag{10}$$

Trivially,

$$\eta x L^{-1} \int_{-\sqrt{L}}^{\sqrt{L}} |M(b+it)H(b+it)| dt \ll \eta x L^{-\frac{1}{2}} \ll \eta x^{1-\varepsilon}, \tag{11}$$

$$\eta^2 x \int_{-\sqrt{L}}^{\sqrt{L}} |M(b+it)H(b+it)| dt \ll \eta^2 x L^{\frac{1}{2}} \ll \eta x^{1-\varepsilon}. \tag{12}$$

Hence

$$I_1 = \frac{\eta}{2\pi i} \int_{b-i\sqrt{L}}^{b+i\sqrt{L}} M(s)H(s) \frac{(c_2L)^{1-s} - (c_1L)^{1-s}}{1-s} x^s ds + O(\eta x^{1-\varepsilon}). \tag{13}$$

By Perron's formula again, we have

$$\begin{aligned}
\frac{\eta}{2\pi i} \int_{b-i\sqrt{L}}^{b+i\sqrt{L}} M(s)H(s) \frac{(c_2L)^{1-s} - (c_1L)^{1-s}}{1-s} x^s ds &= \sum_{\substack{m \sim M \\ h \sim H}} \left(a_m b_h \frac{\eta x}{mh} \right) + O\left(\frac{\eta x^{1+\varepsilon}}{\sqrt{L}}\right) + O\left(\frac{\eta x^{1+\varepsilon}}{MH}\right) \\
&= \sum_{\substack{m \sim M \\ h \sim H}} \left(a_m b_h \frac{\eta x}{mh} \right) + O(\eta x^{1-\varepsilon}).
\end{aligned} \tag{14}$$

By (8), we can easily show that (write $|t| \sim T$)

$$\frac{1-(1-\eta)^s}{s} \ll \min\left(\eta, \frac{1}{T}\right).$$

Thus, by Lemma 4.2 and the essentially same method as in [[26], Lemma 16], we can prove that the number of x such that $I_2 \ll \eta x (\log x)^{-B}$ or $I_3 \ll \eta x (\log x)^{-B}$ fail is $\ll X (\log X)^{-B}$. Now by (5)–(14) and above discussions, Lemma 6.1 is proved. $\square$

Lemma 6.2. Suppose that $NQR L = X$, and N , Q and R satisfy the following condition:

$$X^{\frac{85}{198}} \ll NQ \ll X^{\frac{255}{539}}, \quad X^{\frac{20}{209}} \ll R \ll Q \ll (NQ)^{-\frac{1}{5}} X^{\frac{3}{11}}.$$

Assume further that for $\sqrt{L} \leq |t| \leq 2X$, $N(b+it)Q(b+it) \ll (\log x)^{-B/\varepsilon}$ and $R(b+it) \ll (\log x)^{-B/\varepsilon}$. Then for real numbers $x \in [X, 2X]$, except for $O(X(\log X)^{-B})$ values, we have Then for real numbers $x \in [X, 2X]$, except for $O(X(\log X)^{-B})$ values, we have

$$\sum_{\substack{n \sim N \\ q \sim Q \\ r \sim R}} a_n b_q c_r \left(\sum_{x-\eta x \leq nqr l < x} 1 - \frac{\eta x}{nqr} \right) \ll \eta x (\log x)^{-B}.$$

Proof. The proof is similar to that of Lemma 6.1 where Lemma 4.3 is used. $\square$

The next three lemmas focus on the case $z = p$, or the general case.

Lemma 6.3. Suppose that $X^{\frac{189}{209}} \ll M \ll X^{1-\delta}$, $a_m \geq 0$, and $a_m = 0$ if m has a prime factor $< X^\delta$. Then for real numbers $x \in [X, 2X]$, except for $O(X(\log X)^{-B})$ values, we have

$$\sum_{\substack{x-\eta x \leq mp < x \\ m \sim M}} a_m = \eta \left(1 + O\left(\frac{1}{\log x}\right) \right) \sum_{\substack{x < mp \leq 2x \\ m \sim M}} a_m + O(\eta x (\log x)^{-B}).$$

Proof. The proof is similar to that of Lemma 6.1. Let $F(s) = M(s)H(s)$. By Perron's formula, we have

$$\sum_{\substack{x-\eta x \leq mh < x \\ m \sim M}} a_m \Lambda(h) = \frac{1}{2\pi i} \int_{b-iX}^{b+iX} F(s) \frac{1-(1-\eta)^s}{s} x^s ds + O(x^\varepsilon). \tag{15}$$

We know that

$$\frac{1}{2\pi i} \int_{b-iX}^{b+iX} F(s) \frac{1-(1-\eta)^s}{s} x^s ds = \frac{1}{2\pi i} \int_{b-i(\log X)^{B/\varepsilon}}^{b+i(\log X)^{B/\varepsilon}} F(s) \frac{1-(1-\eta)^s}{s} x^s ds$$

$$\begin{aligned}
& + \frac{1}{2\pi i} \int_{b+i(\log X)^{B/\varepsilon}}^{b+iX} F(s) \frac{1-(1-\eta)^s}{s} x^s ds \\
& + \frac{1}{2\pi i} \int_{b-iX}^{b-i(\log X)^{B/\varepsilon}} F(s) \frac{1-(1-\eta)^s}{s} x^s ds \\
& = I_4 + I_5 + I_6.
\end{aligned} \tag{16}$$

By (4) and (8), we know that

$$\begin{aligned}
I_4 &= \frac{1}{2\pi i} \int_{b-i(\log X)^{B/\varepsilon}}^{b+i(\log X)^{B/\varepsilon}} F(s) \frac{1-(1-\eta)^s}{s} x^s ds = \frac{\eta}{2\pi i} \int_{b-i(\log X)^{B/\varepsilon}}^{b+i(\log X)^{B/\varepsilon}} M(s) \frac{(c_2 H)^{1-s} - (c_1 H)^{1-s}}{1-s} x^s ds \\
& + O\left(\eta x (\log x)^{-2B/\varepsilon} \int_{-(\log X)^{B/\varepsilon}}^{(\log X)^{B/\varepsilon}} |M(b+it)| dt\right) \\
& + O\left(\eta^2 x \int_{-(\log X)^{B/\varepsilon}}^{(\log X)^{B/\varepsilon}} |M(b+it)| dt\right).
\end{aligned} \tag{17}$$

By trivial estimates and Perron's formula again, we have

$$\begin{aligned}
I_4 &= \frac{\eta}{2\pi i} \int_{b-i(\log X)^{B/\varepsilon}}^{b+i(\log X)^{B/\varepsilon}} M(s) \frac{(c_2 H)^{1-s} - (c_1 H)^{1-s}}{1-s} x^s ds + O\left(\eta x (\log x)^{-2B}\right) \\
&= \sum_{m \sim M} \left(a_m \frac{\eta x}{m}\right) + O\left(\eta x (\log x)^{-2B}\right) + O\left(\frac{\eta x (\log x)^2}{(\log X)^{B/\varepsilon}}\right) + O\left(\frac{\eta x (\log x)^2}{M}\right) \\
&= \sum_{m \sim M} \left(a_m \frac{\eta x}{m}\right) + O\left(\eta x (\log x)^{-2B}\right).
\end{aligned} \tag{18}$$

Note that we have $X^\delta \ll H \ll X^{\frac{20}{209}}$. By Lemma 3.1 and the essentially same method as in [[26], Lemma 16], we can prove that the number of x such that $I_5 \ll \eta x (\log x)^{-2B}$ or $I_6 \ll \eta x (\log x)^{-2B}$ fail is $\ll X (\log X)^{-B}$. Now,

$$\begin{aligned}
\sum_{\substack{x-\eta x \leq m h < x \\ m \sim M}} a_m \Lambda(h) &= I_4 + I_5 + I_6 + O(x^\varepsilon) \\
&= \sum_{m \sim M} \left(a_m \frac{\eta x}{m}\right) + O\left(\eta x (\log x)^{-2B}\right),
\end{aligned} \tag{19}$$

$$\begin{aligned}
\sum_{\substack{x-\eta x \leq m p < x \\ m \sim M}} a_m &= \left(1 + O\left(\frac{1}{\log x}\right)\right) \sum_{m \sim M} \left(a_m \frac{\eta x}{m \log(x/m)}\right) + O\left(\eta x (\log x)^{-B}\right), \\
&= \eta \left(1 + O\left(\frac{1}{\log x}\right)\right) \sum_{\substack{x < m p \leq 2x \\ m \sim M}} a_m + O\left(\eta x (\log x)^{-B}\right).
\end{aligned} \tag{20}$$

Lemma 6.3 is proved. $\square$

Lemma 6.4. Suppose that $MHK = X$, $c_k = \Lambda(k)$, $X^{\frac{20}{209}} \ll MH \ll X^{\frac{189}{209}}$, and $a_m, b_h \geq 0$. Suppose that $a_m = 0$ if m has a prime factor $< X^\delta$, and $b_h = 0$ if h has a prime factor $< X^\delta$. If we have

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 \int_T^{2T} |M(b+it)H(b+it)K(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}$$

for $(\log X)^{B/\varepsilon} \leq T \leq X$, then for real numbers $x \in [X, 2X]$, except for $O(X (\log X)^{-B})$ values, we have

$$\sum_{\substack{x-\eta x \leq m h p < x \\ m \sim M \\ h \sim H}} a_m b_h = \eta \left(1 + O\left(\frac{1}{\log x}\right)\right) \sum_{\substack{x < m h p \leq 2x \\ m \sim M \\ h \sim H}} a_m b_h + O\left(\eta x (\log x)^{-B}\right).$$

Proof. The proof is similar to that of Lemma 6.3. $\square$

Lemma 6.5. Suppose that $MN HK = X$, $d_k = \Lambda(k)$, $X^{\frac{20}{209}} \ll MNH \ll X^{\frac{189}{209}}$, and $a_m, b_n, c_h \geq 0$. Suppose that $a_m = 0$ if m has a prime factor $< X^\delta$, $b_n = 0$ if n has a prime factor $< X^\delta$, and $c_h = 0$ if h has a prime factor $< X^\delta$. If we have

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 \int_T^{2T} |M(b+it)N(b+it)H(b+it)K(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}$$

for $(\log X)^{B/\varepsilon} \leq T \leq X$, then for real numbers $x \in [X, 2X]$, except for $O(X (\log X)^{-B})$ values, we have

$$\sum_{\substack{x-\eta x \leq m n h p < x \\ m \sim M \\ n \sim N \\ h \sim H}} a_m b_n c_h = \eta \left(1 + O\left(\frac{1}{\log x}\right)\right) \sum_{\substack{x < m n h p \leq 2x \\ m \sim M \\ n \sim N \\ h \sim H}} a_m b_n c_h + O\left(\eta x (\log x)^{-B}\right).$$

Proof. The proof is similar to that of Lemma 6.3. $\square$

The next two lemmas give asymptotic formulas for specific sieve functions.

Lemma 6.6. *Suppose that $P_0 P_1 P_2 \cdots P_n = X$, $P_n < \cdots < P_2 < P_1 < X^{\frac{1}{2}}$ and $p_i \sim P_i$ for $i \geq 1$. Suppose that $P_1, P_2, \dots, P_n, X^{10^{-1000}}$ can be partitioned into M, H satisfy Lemma 6.1 or N, Q, R satisfy Lemma 6.2, Then for real numbers $x \in [X, 2X]$, except for $O(X(\log X)^{-B})$ values, we have*

$$\sum_{(t_1, \dots, t_n)} S(\mathcal{A}_{p_1 p_2 \cdots p_n}, X^{\frac{20}{209}}) = \eta(1 + o(1)) \sum_{(t_1, \dots, t_n)} S(\mathcal{B}_{p_1 p_2 \cdots p_n}^1, X^{\frac{20}{209}}).$$

Proof. If $P_1, P_2, \dots, P_n, X^{10^{-1000}}$ are partitioned into regions that satisfy

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 \int_T^{2T} |M(b+it)H(b+it)L(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B}$$

or

$$\left(\min\left(\eta, \frac{1}{T}\right)\right)^2 \int_T^{2T} |N(b+it)Q(b+it)R(b+it)L(b+it)|^2 dt \ll \eta^2 (\log x)^{-10B},$$

we use Buchstab's identity to get

$$\sum_{(t_1, \dots, t_n)} S(\mathcal{A}_{p_1 p_2 \cdots p_n}, X^{\frac{20}{209}}) = \sum_{(t_1, \dots, t_n)} S(\mathcal{A}_{p_1 p_2 \cdots p_n}, X^\delta) - \sum_{\substack{(t_1, \dots, t_n) \\ \delta \leq t_{n+1} < \frac{20}{209}}} S(\mathcal{A}_{p_1 p_2 \cdots p_n p_{n+1}}, p_{n+1}). \quad (21)$$

We use Iwaniec's linear sieve (taking $D = X^{10^{-1000}}$, $z = X^\delta$) together with Lemma 6.1 or Lemma 6.2 (depending on the region that $P_1, P_2, \dots, P_n, X^{10^{-1000}}$ are partitioned into) to deal with the first term on the right-hand side of (21). The functions $F(s)$ and $f(s)$ defined by the following differential-difference equation

$$\begin{cases} F(s) = \frac{2e^\gamma}{s}, & f(s) = 0, & 0 < s \leq 2, \\ (sF(s))' = f(s-1), & (sf(s))' = F(s-1), & s \geq 2, \end{cases}$$

give the same value 1 with negligible error $O(\varepsilon^2)$, where γ is the Euler's constant. Hence we have

$$\sum_{(t_1, \dots, t_n)} S(\mathcal{A}_{p_1 p_2 \cdots p_n}, X^\delta) = \eta \sum_{(t_1, \dots, t_n)} S(\mathcal{B}_{p_1 p_2 \cdots p_n}^1, X^\delta) + O\left(\frac{\varepsilon \eta x}{\log x}\right). \quad (22)$$

Let $M = \frac{X}{P_{n+1}}$. Clearly we have $X^{\frac{189}{209}} \ll M \ll X^{1-\delta}$. By Lemma 6.3, we have

$$\sum_{\substack{(t_1, \dots, t_n) \\ \delta \leq t_{n+1} < \frac{20}{209}}} S(\mathcal{A}_{p_1 p_2 \cdots p_n p_{n+1}}, p_{n+1}) = \eta \sum_{\substack{(t_1, \dots, t_n) \\ \delta \leq t_{n+1} < \frac{20}{209}}} S(\mathcal{B}_{p_1 p_2 \cdots p_n p_{n+1}}^1, p_{n+1}) + O\left(\frac{\varepsilon \eta x}{\log x}\right). \quad (23)$$

Now, Lemma 6.6 is proved by (21)–(23) and another application of Buchstab's identity

$$\sum_{(t_1, \dots, t_n)} S(\mathcal{B}_{p_1 p_2 \cdots p_n}^1, X^\delta) - \sum_{\substack{(t_1, \dots, t_n) \\ \delta \leq t_{n+1} < \frac{20}{209}}} S(\mathcal{B}_{p_1 p_2 \cdots p_n p_{n+1}}^1, p_{n+1}) = \sum_{(t_1, \dots, t_n)} S(\mathcal{B}_{p_1 p_2 \cdots p_n}^1, X^{\frac{20}{209}}). \quad (24)$$

We remark that in many places, we ignore the additional variable $X^{10^{-1000}}$ when applying this lemma since it only produces an extremely small difference (less than 10^{-10}). $\square$

Lemma 6.7. *Suppose that $P_0 P_1 P_2 P_3 = X$, $P_3 < P_2 < P_1 < X^{\frac{1}{2}}$ and $p_i \sim P_i$ for $i \geq 1$. Suppose that P_1, P_2, P_3 satisfy one of the conditions:*

- (1) $X^{\frac{339}{836}} \ll P_1 P_2 \ll X^{\frac{59}{143}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-\frac{35}{23}} X^{\frac{196}{253}}$;
- (2) $X^{\frac{59}{143}} \ll P_1 P_2 \ll X^{\frac{239}{572}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-1} X^{\frac{80}{143}}$;
- (3) $X^{\frac{239}{572}} \ll P_1 P_2 \ll X^{\frac{47}{110}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-\frac{41}{27}} X^{\frac{461}{594}}$;
- (4) $X^{\frac{47}{110}} \ll P_1 P_2 \ll X^{\frac{377}{836}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-\frac{4}{3}} X^{\frac{23}{33}}$;
- (5) $X^{\frac{379}{836}} \ll P_1 P_2 \ll X^{\frac{5}{11}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-\frac{1}{5}} X^{\frac{3}{11}}$;
- (6) $X^{\frac{5}{11}} \ll P_1 P_2 \ll X^{\frac{31}{66}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll X^{\frac{2}{11}}$;
- (7) $X^{\frac{31}{66}} \ll P_1 P_2 \ll X^{\frac{21}{44}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{\frac{1}{7}} X^{\frac{2}{11}}$;
- (8) $X^{\frac{21}{44}} \ll P_1 P_2 \ll X^{\frac{58}{121}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-1} X^{\frac{8}{11}}$;
- (9) $X^{\frac{58}{121}} \ll P_1 P_2 \ll X^{\frac{243}{506}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-4} X^{\frac{23}{11}}$;
- (10) $X^{\frac{243}{506}} \ll P_1 P_2 \ll X^{\frac{1281}{2662}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-\frac{1}{6}} X^{\frac{1}{4}}$;
- (11) $X^{\frac{1281}{2662}} \ll P_1 P_2 \ll X^{\frac{85}{176}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-2} X^{\frac{137}{121}}$;
- (12) $X^{\frac{85}{176}} \ll P_1 P_2 \ll X^{\frac{139}{286}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-\frac{70}{11}} X^{\frac{392}{121}}$;
- (13) $X^{\frac{139}{286}} \ll P_1 P_2 \ll X^{\frac{417}{836}}$, $X^{\frac{20}{209}} \ll P_3 \ll P_2 \ll (P_1 P_2)^{-4} X^{\frac{23}{11}}$.

Then

$$\sum_{(t_1, t_2, t_3)} S(\mathcal{A}_{p_1 p_2 p_3}, p_3)$$

has an asymptotic formula of the form (3).

Proof. Let $N(s) = P_1(s)$, $Q(s) = P_2(s)$ and $R(s) = P_3(s)$. Combining Lemmas 3.3 and 5.3 and use Lemmas 6.5 and 5.1, Lemma 6.7 is proved. $\square$

7. THE FINAL DECOMPOSITION

In this section, sets $\mathcal{A}$, $\mathcal{B}^1$ and $\mathcal{B}^2$ are defined respectively. Let $\omega(u)$ denote the Buchstab function determined by the following differential-difference equation

$$\begin{cases} \omega(u) = \frac{1}{u}, & 1 \leq u \leq 2, \\ (u\omega(u))' = \omega(u-1), & u \geq 2. \end{cases}$$

Moreover, we have the upper and lower bounds for $\omega(u)$:

$$\begin{aligned} \omega(u) \geq \omega_0(u) &= \begin{cases} \frac{1}{u}, & 1 \leq u < 2, \\ \frac{1}{1+\log(u-1)}, & 2 \leq u < 3, \\ \frac{1+\log(u-1)}{u} + \frac{1}{u} \int_2^{u-1} \frac{\log(t-1)}{t} dt \geq 0.5607, & 3 \leq u < 4, \\ 0.5612, & u \geq 4, \end{cases} \\ \omega(u) \leq \omega_1(u) &= \begin{cases} \frac{1}{u}, & 1 \leq u < 2, \\ \frac{1}{1+\log(u-1)}, & 2 \leq u < 3, \\ \frac{1+\log(u-1)}{u} + \frac{1}{u} \int_2^{u-1} \frac{\log(t-1)}{t} dt \leq 0.5644, & 3 \leq u < 4, \\ 0.5617, & u \geq 4. \end{cases} \end{aligned}$$

We shall use $\omega_0(u)$ and $\omega_1(u)$ to give numerical bounds for some sieve functions discussed below.

By Prime Number Theorem with Vinogradov's error term and the inductive arguments in [[14], Chapter A.2], we know that, for sufficiently large z ,

$$S(\mathcal{B}^1, z) = \sum_{\substack{a \in \mathcal{B}^1 \\ (a, P(z))=1}} 1 = (1 + o(1)) \frac{x}{\log z} \omega\left(\frac{\log x}{\log z}\right), \quad (25)$$

$$S(\mathcal{B}^2, z) = \sum_{\substack{a \in \mathcal{B}^2 \\ (a, P(z))=1}} 1 = (1 + o(1)) \frac{\eta_1 x}{\log z} \omega\left(\frac{\log x}{\log z}\right), \quad (26)$$

and we expect that the similar relation also holds for $S(\mathcal{A}, z)$:

$$S(\mathcal{A}, z) = \sum_{\substack{a \in \mathcal{A} \\ (a, P(z))=1}} 1 = (1 + o(1)) \frac{\eta x}{\log z} \omega\left(\frac{\log x}{\log z}\right). \quad (27)$$

If (3) holds for $S(\mathcal{A}, z)$, then we can deduce (27) easily from (25) or (26). Otherwise we must drop this $S(\mathcal{A}, z)$. We define the *loss* from this term by the size of corresponding $S(\mathcal{B}^i, z)$:

$$S(\mathcal{B}^1, z) = (\text{loss} + o(1)) \frac{x}{\log x} \quad \text{or} \quad S(\mathcal{B}^2, z) = (\text{loss} + o(1)) \frac{\eta_1 x}{\log x}. \quad (28)$$

We note that we can only drop positive parts and the total loss of the dropped parts must be less than 1.

Beginning with (2), we can easily give asymptotic formulas for S_1 and S_2 by Lemma 6.6. Before estimating S_3 , we first define regions U_{01} – U_{04} and V as

$$\begin{aligned} U_{01}(t_1, t_2) &:= \{(t_1, t_2) \in U_{0101} \cup U_{0102} \cup \dots \cup U_{0109}, \\ &\quad \frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min\left(t_1, \frac{1}{2}(1-t_1)\right)\}, \\ V(t_1, t_2) &:= \{(t_1, t_2) \in V_0 \cup V_{11} \cup V_{12}, \\ &\quad \frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min\left(t_1, \frac{1}{2}(1-t_1)\right)\}, \\ U_{02}(t_1, t_2) &:= \{(t_1, t_2) \in U_{0201} \cup U_{0202} \cup \dots \cup U_{0216}, (t_1, t_2) \notin U_{01} \cup V, \\ &\quad \frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min\left(t_1, \frac{1}{2}(1-t_1)\right)\}, \\ U_{03}(t_1, t_2) &:= \{(t_1, t_2) \in U_{0301} \cup U_{0302} \cup U_{0303}, (t_1, t_2) \notin U_{01} \cup V \cup U_{02}, \\ &\quad \frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min\left(t_1, \frac{1}{2}(1-t_1)\right)\}, \\ U_{04}(t_1, t_2) &:= \{(t_1, t_2) \in U_{0401} \cup U_{0402}, (t_1, t_2) \notin U_{01} \cup V \cup U_{02} \cup U_{03}, \end{aligned}$$

$$\frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min \left(t_1, \frac{1}{2}(1-t_1) \right) \Big\},$$

where

$$\begin{aligned} U_{0101}(t_1, t_2) &:= \left\{ t_1 + t_2 \leq \frac{172}{319}, \frac{21}{121} \leq t_2, 29t_1 - t_2 \leq \frac{109}{11}, \frac{63}{209} \leq t_1, 29t_2 - t_1 \leq \frac{67}{11}, \frac{63}{11} \leq 12t_1 + 11t_2 \right\}, \\ U_{0102}(t_1, t_2) &:= \left\{ t_1 + t_2 \leq \frac{19}{33}, 29t_1 + 19t_2 \leq \frac{323}{22}, \frac{28}{33} \leq 2t_1 + t_2, 2t_1 + 11t_2 \leq \frac{30}{11}, \frac{315}{11} \leq 58t_1 + 49t_2 \right\}, \\ U_{0103}(t_1, t_2) &:= \left\{ t_1 + t_2 \leq \frac{137}{242}, \frac{21}{110} \leq t_2, 6t_1 + t_2 \leq \frac{24}{11}, \frac{3}{11} \leq t_1 \leq \frac{4}{11}, t_1 + 8t_2 \leq \frac{25}{11}, \frac{63}{22} \leq 6t_1 + 5t_2 \right\}, \\ U_{0104}(t_1, t_2) &:= \left\{ t_1 + t_2 \leq \frac{146}{209}, \frac{21}{121} \leq t_2, 12t_1 + t_2 \leq \frac{69}{11}, \frac{147}{319} \leq t_1, 19t_2 - t_1 \leq \frac{41}{11}, \frac{84}{11} \leq 12t_1 + 11t_2 \right\}, \\ U_{0105}(t_1, t_2) &:= \left\{ 2t_1 + t_2 \leq \frac{38}{33}, 58t_1 + 9t_2 \leq \frac{323}{11}, \frac{14}{33} \leq t_1, t_1 + 5t_2 \leq \frac{15}{11}, \frac{315}{22} \leq 29t_1 + 10t_2 \right\}, \\ U_{0106}(t_1, t_2) &:= \left\{ \frac{7}{11} \leq t_1 + t_2 \leq \frac{8}{11}, \frac{21}{110} \leq t_2, 6t_1 + t_2 \leq \frac{69}{22}, \frac{105}{242} \leq t_1, 7t_2 - t_1 \leq \frac{14}{11}, \frac{42}{11} \leq 6t_1 + 5t_2 \right\}, \\ U_{0107}(t_1, t_2) &:= \left\{ t_1 + t_2 \leq \frac{137}{242}, 35t_1 + 23t_2 \leq \frac{196}{11}, \frac{105}{121} \leq 2t_1 + t_2, 2t_1 + 13t_2 \leq 3, \frac{378}{11} \leq 70t_1 + 59t_2 \right\}, \\ U_{0108}(t_1, t_2) &:= \left\{ t_1 + t_2 \leq \frac{80}{143}, 41t_1 + 27t_2 \leq \frac{461}{22}, \frac{126}{143} \leq 2t_1 + t_2, 2t_1 + 15t_2 \leq \frac{35}{11}, \frac{441}{11} \leq 82t_1 + 69t_2 \right\}, \\ U_{0109}(t_1, t_2) &:= \left\{ 2t_1 + t_2 \leq \frac{137}{121}, 70t_1 + 11t_2 \leq \frac{392}{11}, \frac{105}{242} \leq t_1, t_1 + 6t_2 \leq \frac{3}{2}, \frac{189}{11} \leq 35t_1 + 12t_2 \right\}, \\ V_0(t_1, t_2) &:= \left\{ t_2 > \frac{1}{11}, 2t_2 - \frac{1}{11} < t_1 < \frac{2}{11} \right\}, \\ V_{11}(t_1, t_2) &:= \left\{ |4t_1 + 3t_2 - 2| < \frac{1}{11}, \frac{1}{11} < t_2 < \frac{2}{11} \right\}, \\ V_{11}(t_1, t_2) &:= \left\{ |4t_1 + t_2 - 2| < \frac{1}{11}, \frac{1}{11} < t_2 < \frac{2}{11} \right\}, \\ U_{0201}(t_1, t_2) &:= \left\{ \frac{339}{836} \leq t_1 + t_2 \leq \frac{59}{143}, \frac{20}{209} \leq t_2 \leq -\frac{35}{23}(t_1 + t_2) + \frac{196}{253} \right\}, \\ U_{0202}(t_1, t_2) &:= \left\{ \frac{59}{143} \leq t_1 + t_2 \leq \frac{239}{572}, \frac{20}{209} \leq t_2 \leq -(t_1 + t_2) + \frac{80}{143} \right\}, \\ U_{0203}(t_1, t_2) &:= \left\{ \frac{239}{572} \leq t_1 + t_2 \leq \frac{47}{110}, \frac{20}{209} \leq t_2 \leq -\frac{41}{27}(t_1 + t_2) + \frac{461}{594} \right\}, \\ U_{0204}(t_1, t_2) &:= \left\{ \frac{47}{110} \leq t_1 + t_2 \leq \frac{377}{836}, \frac{20}{209} \leq t_2 \leq -\frac{4}{3}(t_1 + t_2) + \frac{23}{33} \right\}, \\ U_{0205}(t_1, t_2) &:= \left\{ \frac{379}{836} \leq t_1 + t_2 \leq \frac{5}{11}, \frac{20}{209} \leq t_2 \leq -\frac{1}{5}(t_1 + t_2) + \frac{3}{11} \right\}, \\ U_{0206}(t_1, t_2) &:= \left\{ \frac{5}{11} \leq t_1 + t_2 \leq \frac{31}{66}, \frac{20}{209} \leq t_2 \leq \frac{2}{11} \right\}, \\ U_{0207}(t_1, t_2) &:= \left\{ \frac{31}{66} \leq t_1 + t_2 \leq \frac{21}{44}, \frac{20}{209} \leq t_2 \leq \frac{1}{7}(t_1 + t_2) + \frac{2}{11} \right\}, \\ U_{0208}(t_1, t_2) &:= \left\{ \frac{21}{44} \leq t_1 + t_2 \leq \frac{58}{121}, \frac{20}{209} \leq t_2 \leq -(t_1 + t_2) + \frac{8}{11} \right\}, \\ U_{0209}(t_1, t_2) &:= \left\{ \frac{58}{121} \leq t_1 + t_2 \leq \frac{243}{506}, \frac{20}{209} \leq t_2 \leq -4(t_1 + t_2) + \frac{23}{11} \right\}, \\ U_{0210}(t_1, t_2) &:= \left\{ \frac{243}{506} \leq t_1 + t_2 \leq \frac{1281}{2662}, \frac{20}{209} \leq t_2 \leq -\frac{1}{6}(t_1 + t_2) + \frac{1}{4} \right\}, \\ U_{0211}(t_1, t_2) &:= \left\{ \frac{1281}{2662} \leq t_1 + t_2 \leq \frac{85}{176}, \frac{20}{209} \leq t_2 \leq -2(t_1 + t_2) + \frac{137}{121} \right\}, \\ U_{0212}(t_1, t_2) &:= \left\{ \frac{85}{176} \leq t_1 + t_2 \leq \frac{139}{286}, \frac{20}{209} \leq t_2 \leq -\frac{70}{11}(t_1 + t_2) + \frac{392}{121} \right\}, \\ U_{0213}(t_1, t_2) &:= \left\{ \frac{139}{286} \leq t_1 + t_2 \leq \frac{417}{836}, \frac{20}{209} \leq t_2 \leq -4(t_1 + t_2) + \frac{23}{11} \right\}, \\ U_{0214}(t_1, t_2) &:= \left\{ t_1 + (t_2 + t_2) \leq \frac{172}{319}, \frac{21}{121} \leq \left(t_2 + \frac{20}{209} \right), 29t_1 - \left(t_2 + \frac{20}{209} \right) \leq \frac{109}{11}, \frac{63}{209} \leq t_1, \right. \end{aligned}$$

$$\begin{aligned}
& 29(t_2 + t_2) - t_1 \leq \frac{67}{11}, \frac{63}{11} \leq 12t_1 + 11 \left(t_2 + \frac{20}{209} \right) \Big\}, \\
U_{0215}(t_1, t_2) &:= \left\{ t_1 + (t_2 + t_2) \leq \frac{137}{242}, \frac{21}{110} \leq \left(t_2 + \frac{20}{209} \right), 6t_1 + (t_2 + t_2) \leq \frac{24}{11}, \frac{3}{11} \leq t_1 \leq \frac{4}{11}, \right. \\
& \quad \left. t_1 + 8(t_2 + t_2) \leq \frac{25}{11}, \frac{63}{22} \leq 6t_1 + 5 \left(t_2 + \frac{20}{209} \right) \right\}, \\
U_{0216}(t_1, t_2) &:= \left\{ t_1 + (t_2 + t_2) \leq \frac{146}{209}, \frac{21}{121} \leq \left(t_2 + \frac{20}{209} \right), 12t_1 + (t_2 + t_2) \leq \frac{69}{11}, \frac{147}{319} \leq t_1, \right. \\
& \quad \left. 19(t_2 + t_2) - t_1 \leq \frac{41}{11}, \frac{84}{11} \leq 12t_1 + 11 \left(t_2 + \frac{20}{209} \right) \right\}, \\
U_{0301}(t_1, t_2) &:= \left\{ 2(t_1 + t_2) + t_2 \leq \frac{23}{22}, 4(t_1 + t_2) + 6t_2 \leq \frac{67}{22}, 4t_2 \leq \frac{25}{22} \right\}, \\
U_{0302}(t_1, t_2) &:= \left\{ 2(t_2 + t_2) + t_1 \leq \frac{23}{22}, 4(t_2 + t_2) + 6t_1 \leq \frac{67}{22}, 4t_1 \leq \frac{25}{22} \right\}, \\
U_{0303}(t_1, t_2) &:= \left\{ \frac{85}{198} \leq (t_1 + t_2) \leq \frac{255}{539}, t_2 \leq -\frac{1}{5}(t_1 + t_2) + \frac{3}{11} \right\}, \\
U_{0401}(t_1, t_2) &:= \left\{ 2t_1 + t_2 \leq \frac{23}{22}, 4t_1 + 6t_2 \leq \frac{67}{22}, 4t_2 \leq \frac{25}{22}, \right. \\
& \quad \left. 2(1 - t_1 - t_2) + t_2 \leq \frac{23}{22}, 4(1 - t_1 - t_2) + 6t_2 \leq \frac{67}{22} \right\}, \\
U_{0402}(t_1, t_2) &:= \left\{ t_1 \leq \frac{23}{44}, t_2 \leq \frac{21}{176}, (1 - t_1 - t_2) \leq \frac{23}{44} \right\},
\end{aligned}$$

and let $U_0 = U_{01} \cup V \cup U_{02} \cup U_{03} \cup U_{04}$. Clearly U_{01} corresponds to Lemma 3.2, V corresponds to Lemma 5.2, U_{02} corresponds to Lemmas 3.3–3.4 and U_{03} corresponds to Lemmas 4.2–4.3. Then we have

$$\begin{aligned}
S_3 &= \sum_{(t_1, t_2) \in U_{01} \cup V} S(\mathcal{A}_{p_1 p_2}, p_2) + \sum_{(t_1, t_2) \in U_{02}} S(\mathcal{A}_{p_1 p_2}, p_2) \\
&+ \sum_{(t_1, t_2) \in U_{03}} S(\mathcal{A}_{p_1 p_2}, p_2) + \sum_{(t_1, t_2) \in U_{04}} S(\mathcal{A}_{p_1 p_2}, p_2) \\
&+ \sum_{(t_1, t_2) \notin U_0} S(\mathcal{A}_{p_1 p_2}, p_2) \\
&= S_{301} + S_{302} + S_{303} + S_{304} + S_{305}.
\end{aligned} \tag{29}$$

For S_{301} , by Lemmas 3.2, 6.4, 5.1 and 5.2, we can give an asymptotic formula.

For S_{302} , we can apply Buchstab's identity again to get

$$\begin{aligned}
S_{302} &= \sum_{(t_1, t_2) \in U_{02}} S(\mathcal{A}_{p_1 p_2}, p_2) \\
&= \sum_{(t_1, t_2) \in U_{02}} S\left(\mathcal{A}_{p_1 p_2}, X^{\frac{20}{209}}\right) - \sum_{\substack{(t_1, t_2) \in U_{02} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1 - t_1 - t_2))}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3).
\end{aligned} \tag{30}$$

By Lemma 6.6 we can give an asymptotic formula for

$$\sum_{(t_1, t_2) \in U_{02}} S\left(\mathcal{A}_{p_1 p_2}, X^{\frac{20}{209}}\right). \tag{31}$$

By Lemma 6.7 we can give an asymptotic formula for

$$\sum_{\substack{(t_1, t_2) \in U_{02} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1 - t_1 - t_2))}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3). \tag{32}$$

Combining (30)–(32), we can give an asymptotic formula for S_{302} .

Before decomposing S_{303} , we first make a definition: we say a set of variables $(t_1, \dots, t_n)$ is *good* if the variables can be partitioned into $(t_i, h) \in U_{01}$, $(h, t_i) \in U_{01}$ or $(m, h) \in V$. Now we can apply Buchstab's identity twice more to get

$$\begin{aligned}
S_{303} &= \sum_{(t_1, t_2) \in U_{03}} S(\mathcal{A}_{p_1 p_2}, p_2) \\
&= \sum_{(t_1, t_2) \in U_{03}} S\left(\mathcal{A}_{p_1 p_2}, X^{\frac{20}{209}}\right) - \sum_{\substack{(t_1, t_2) \in U_{03} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1 - t_1 - t_2)) \\ (t_1, t_2, t_3) \text{ is good}}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3)
\end{aligned}$$

$$\begin{aligned}
& - \sum_{\substack{(t_1, t_2) \in U_{03} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good}}} S(\mathcal{A}_{p_1 p_2 p_3}, X^{\frac{20}{209}}) + \sum_{\substack{(t_1, t_2) \in U_{03} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good} \\ \frac{20}{209} \leq t_4 < \min(t_3, \frac{1}{2}(1-t_1-t_2-t_3)) \\ (t_1, t_2, t_3, t_4) \text{ is good}}} S(\mathcal{A}_{p_1 p_2 p_3 p_4}, p_4) \\
& + \sum_{\substack{(t_1, t_2) \in U_{03} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good} \\ \frac{20}{209} \leq t_4 < \min(t_3, \frac{1}{2}(1-t_1-t_2-t_3)) \\ (t_1, t_2, t_3, t_4) \text{ is not good}}} S(\mathcal{A}_{p_1 p_2 p_3 p_4}, p_4) \\
& = S_{3031} - S_{3032} - S_{3033} + S_{3034} + S_{3035}.
\end{aligned} \tag{33}$$

By Lemma 6.6 we can give asymptotic formulas for

$$S_{3031} = \sum_{(t_1, t_2) \in U_{03}} S(\mathcal{A}_{p_1 p_2}, X^{\frac{20}{209}}) \quad \text{and} \quad S_{3033} = \sum_{\substack{(t_1, t_2) \in U_{03} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good}}} S(\mathcal{A}_{p_1 p_2 p_3}, X^{\frac{20}{209}}). \tag{34}$$

By Lemmas 3.2, 6.5, 5.1 and 5.2, we can give asymptotic formulas for

$$S_{3032} = \sum_{\substack{(t_1, t_2) \in U_{03} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is good}}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3) \quad \text{and} \quad S_{3034} = \sum_{\substack{(t_1, t_2) \in U_{03} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good} \\ \frac{20}{209} \leq t_4 < \min(t_3, \frac{1}{2}(1-t_1-t_2-t_3)) \\ (t_1, t_2, t_3, t_4) \text{ is good}}} S(\mathcal{A}_{p_1 p_2 p_3 p_4}, p_4). \tag{35}$$

For the remaining S_{3035} , we can simply discard the whole sum, leading to a loss of size

$$\int_{(t_1, t_2, t_3, t_4) \in R_1} \frac{\omega_1 \left(\frac{1-t_1-t_2-t_3-t_4}{t_4} \right)}{t_1 t_2 t_3 t_4^2} dt_4 dt_3 dt_2 dt_1 < 0.01, \tag{36}$$

where

$$\begin{aligned}
R_1(t_1, t_2, t_3, t_4) := & \left\{ (t_1, t_2) \in U_{03}, \frac{20}{209} \leq t_3 < \min \left(t_2, \frac{1}{2}(1-t_1-t_2) \right), \right. \\
& (t_1, t_2, t_3) \text{ is not good}, \\
& \frac{20}{209} \leq t_4 < \min \left(t_3, \frac{1}{2}(1-t_1-t_2-t_3) \right), \\
& (t_1, t_2, t_3, t_4) \text{ is not good}, \\
& \left. \frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min \left(t_1, \frac{1}{2}(1-t_1) \right) \right\}.
\end{aligned}$$

We note that further decompositions on S_{3035} are possible, but we do not consider them here.

One of the most important devices we used in the present paper is the *variable role-reversal*, and we shall use this to deal with S_{304} . By Buchstab's identity, we have

$$\begin{aligned}
S_{304} &= \sum_{(t_1, t_2) \in U_{04}} S(\mathcal{A}_{p_1 p_2}, p_2) \\
&= \sum_{(t_1, t_2) \in U_{04}} S(\mathcal{A}_{p_1 p_2}, X^{\frac{20}{209}}) - \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is good}}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3) \\
&\quad - \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good}}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3) \\
&= S_{3041} - S_{3042} - S_{3043}.
\end{aligned} \tag{37}$$

By Lemma 6.6 we can give an asymptotic formula for

$$S_{3041} = \sum_{(t_1, t_2) \in U_{04}} S(\mathcal{A}_{p_1 p_2}, X^{\frac{20}{209}}). \tag{38}$$

By Lemmas 3.2, 6.5, 5.1 and 5.2, we can give asymptotic formulas for

$$S_{3042} = \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is good}}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3). \tag{39}$$

Note that S_{3043} counts numbers of the form $p_1 p_2 p_3 \beta$, where the implicit variable β satisfies $\beta \sim X^{1-t_1-t_2-t_3}$ and $(\beta, P(p_3)) = 1$. Now, we reverse the roles of variables β and p_1 , making p_1 the implicit one. In this way, S_{3043} can be rewritten as

$$S_{3043} = \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good}}} S\left(\mathcal{A}_{\beta p_2 p_3}, \left(\frac{2X}{\beta p_2 p_3}\right)^{\frac{1}{2}}\right). \quad (40)$$

Using Buchstab's identity, we have

$$\begin{aligned} S_{3043} &= \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good}}} S\left(\mathcal{A}_{\beta p_2 p_3}, \left(\frac{2X}{\beta p_2 p_3}\right)^{\frac{1}{2}}\right) \\ &= \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good}}} S\left(\mathcal{A}_{\beta p_2 p_3}, X^{\frac{20}{209}}\right) \\ &\quad - \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good} \\ \frac{20}{209} \leq t_4 < \frac{1}{2}t_1 \\ (1-t_1-t_2-t_3, t_2, t_3, t_4) \text{ is good}}} S(\mathcal{A}_{\beta p_2 p_3 p_4}, p_4) \\ &\quad - \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good} \\ \frac{20}{209} \leq t_4 < \frac{1}{2}t_1 \\ (1-t_1-t_2-t_3, t_2, t_3, t_4) \text{ is not good}}} S(\mathcal{A}_{\beta p_2 p_3 p_4}, p_4) \\ &= S_{30431} - S_{30432} - S_{30433}. \end{aligned} \quad (41)$$

By Lemma 6.6 and the definition of U_{04} , we can give an asymptotic formula for

$$S_{30431} = \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good}}} S\left(\mathcal{A}_{\beta p_2 p_3}, X^{\frac{20}{209}}\right). \quad (42)$$

By Lemmas 3.2, 6.5, 5.1 and 5.2, we can give asymptotic formulas for

$$S_{30432} = \sum_{\substack{(t_1, t_2) \in U_{04} \\ \frac{20}{209} \leq t_3 < \min(t_2, \frac{1}{2}(1-t_1-t_2)) \\ (t_1, t_2, t_3) \text{ is not good} \\ \frac{20}{209} \leq t_4 < \frac{1}{2}t_1 \\ (1-t_1-t_2-t_3, t_2, t_3, t_4) \text{ is good}}} S(\mathcal{A}_{\beta p_2 p_3 p_4}, p_4). \quad (43)$$

We discard the whole sum S_{30433} , leading to a loss of size

$$\int_{(t_1, t_2, t_3, t_4) \in R_2} \frac{\omega_1\left(\frac{t_1-t_4}{t_4}\right) \omega_1\left(\frac{1-t_1-t_2-t_3}{t_3}\right)}{t_2 t_3^2 t_4^2} dt_4 dt_3 dt_2 dt_1 < 0.02, \quad (44)$$

where

$$\begin{aligned} R_2(t_1, t_2, t_3, t_4) := & \left\{ (t_1, t_2) \in U_{04}, \frac{20}{209} \leq t_3 < \min\left(t_2, \frac{1}{2}(1-t_1-t_2)\right), \right. \\ & (t_1, t_2, t_3) \text{ is not good,} \\ & \frac{20}{209} \leq t_4 < \frac{1}{2}t_1, \\ & (1-t_1-t_2-t_3, t_2, t_3, t_4) \text{ is not good,} \\ & \left. \frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min\left(t_1, \frac{1}{2}(1-t_1)\right) \right\}. \end{aligned}$$

We remark that if we do not use the variable role-reversal, the loss from discarding the whole S_{304} is larger than 0.2.

For the remaining S_{305} , discarding the whole sum lead to a two-dimensional loss of

$$\int_{\frac{20}{209}}^{\frac{1}{2}} \int_{\frac{20}{209}}^{\min(t_1, \frac{1-t_1}{2})} \mathbb{1}_{(t_1, t_2) \notin U_0} \frac{\omega\left(\frac{1-t_1-t_2}{t_2}\right)}{t_1 t_2^2} dt_2 dt_1 < 0.8, \quad (45)$$

However, we can use Buchstab's identity in reverse to get asymptotic formulas for some almost-primes counted. When $t_2 < \frac{1}{2}(1 - t_1 - t_2)$, we have

$$\sum_{\substack{(t_1, t_2) \notin U_0 \\ t_2 < \frac{1}{2}(1 - t_1 - t_2)}} S(\mathcal{A}_{p_1 p_2}, p_2) = \sum_{\substack{(t_1, t_2) \notin U_0 \\ t_2 < \frac{1}{2}(1 - t_1 - t_2)}} S\left(\mathcal{A}_{p_1 p_2}, \left(\frac{2X}{p_1 p_2}\right)^{\frac{1}{2}}\right) + \sum_{\substack{(t_1, t_2) \notin U_0 \\ t_2 < t_3 < \frac{1}{2}(1 - t_1 - t_2)}} S(\mathcal{A}_{p_1 p_2 p_3}, p_3). \quad (46)$$

This can be seen as a "reversed" application of Buchstab's identity. The first term on the right-hand side counts primes, while the second term counts almost-primes. We cannot give an asymptotic formula for the first term on the right-hand side, but we can give asymptotic formulas for part of the second term if (t_1, t_2, t_3) is good. We can then subtract the size of this part from the loss of S_{305} . The savings from this part are larger than

$$\int_{(t_1, t_2, t_3) \in R_3} \frac{\omega\left(\frac{1 - t_1 - t_2 - t_3}{t_3}\right)}{t_1 t_2 t_3^2} dt_3 dt_2 dt_1 > 0.03, \quad (47)$$

where

$$R_3(t_1, t_2, t_3) := \left\{ (t_1, t_2) \notin U_0, t_2 < t_3 < \frac{1}{2}(1 - t_1 - t_2), (t_1, t_2, t_3) \text{ is good}, \right. \\ \left. \frac{20}{209} \leq t_1 < \frac{1}{2}, \frac{20}{209} \leq t_2 < \min\left(t_1, \frac{1}{2}(1 - t_1)\right) \right\}.$$

Finally, by (1)–(47) we have

$$\pi(x + \eta x) - \pi(x) = S\left(\mathcal{A}, (2X)^{\frac{1}{2}}\right) \geq (1 - 0.01 - 0.02 - 0.8 + 0.03) \frac{\eta x}{\log x} = 0.2 \frac{\eta x}{\log x},$$

and the proof of Theorem 1.1 is completed.

8. APPLICATIONS OF THEOREM 1.1

Clearly our Theorem 1.1 has many interesting applications. The following application of our Theorem 1.1 is about Goldbach numbers (sum of two primes) in short intervals. By combining our Theorem 1.1 with the main theorem proved in [31], we can easily deduce the following theorem.

Theorem 8.1. *The interval $[X, X + X^{\frac{13}{550}}]$ contains Goldbach numbers. That is,*

$$g_{n+1} - g_n \ll g_n^{\frac{13}{550}},$$

where g_n denote the n -th Goldbach number.

If we focus on Maillet numbers (difference of two primes) instead of Goldbach numbers in short intervals, Pintz [41] improved this exponent to any $\varepsilon > 0$.

Another application of our Theorem 1.1 is about prime values of the integer parts of real sequences, improving the previous result of the author [30] by adding one more term on both of the sequences $[p^k \alpha]$ and $[(p\alpha)^k]$.

Theorem 8.2. *For almost all $\alpha > 0$, both of the following statements are true:*

- (1) *Infinitely often $p, [p\alpha], [p^2\alpha], \dots, [p^{22}\alpha]$ are all prime.*
- (2) *Infinitely often $p, [p\alpha], [(p\alpha)^2], \dots, [(p\alpha)^{22}]$ are all prime.*

The proof of Theorem 8.2 can be done by replacing [[13], Lemma 4] by a variant of our Theorem 1.1. In this way, the ratio $\frac{20}{19}$ in [[13], Theorem 4] can be improved to $\frac{22}{21}$.

The last application of our Theorem 1.1 is about the Three Primes Theorem with small prime solutions, improving the previous result of the author [30] by reducing the exponent $\frac{11}{420}$ to $\frac{1}{40}$.

Theorem 8.3. *Let Y denote a sufficiently large odd integer. The equation*

$$Y = p_1 + p_2 + p_3, \quad p_1 \leq Y^{\frac{1}{40}}$$

is solvable.

We define u_0, u_1, v_0, v_1 as in [5]. Note that we have $u_0 > 0.2$ by Theorem 1.1, and we can show that $u_1 < 2.94$ by similar arguments as in [5]. Using the vector sieve together with the same bounds for v_0 and v_1 as in [5], the proof of our Theorem 8.3 is essentially the same as the proof of [5], Theorem].

9. REMARK

Recently the author has proved Theorem 1.1 with an interval $[n - n^{\frac{1}{24} + \varepsilon}, n]$.

29

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